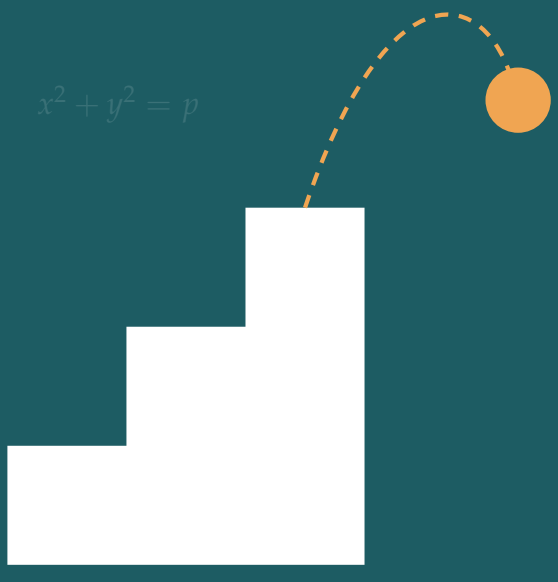
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP II AND III

2020 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP II AND III

2020 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2020 Worked Solutions

Worked solutions to all 24 questions of the 2020 STEP papers (STEP II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP II 2020, Question 1

Pure • Integration by Substitution

STEP II 2020, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Each integrand is a product of powers of two linear factors whose exponents add up to -2. For such integrands the right substitution is the *ratio* of the two linear factors (or its square root): it absorbs both brackets at once, and what is left is a simple power or a standard arctangent. Part (i) shows this happening with the given substitution, part (ii) asks you to copy the trick, and part (iii) asks you to spot it for yourself.

Solution

Part (i)

With $x = \dfrac{1}{1-u}$ we have

$$x - 1 = \frac{1-(1-u)}{1-u} = \frac{u}{1-u}, \qquad \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{(1-u)^2}.$$

As u runs over $(0,1)$, x runs over $(1,\infty)$, and $u = 1 - \frac{1}{x} = \frac{x-1}{x}$. All the quantities under square roots are positive because $0 < u < 1$, so

$$x^{3/2}(x-1)^{1/2} = \frac{1}{(1-u)^{3/2}} \cdot \frac{u^{1/2}}{(1-u)^{1/2}} = \frac{u^{1/2}}{(1-u)^2}.$$

The factors of $(1-u)^2$ now cancel exactly against $\mathrm{d}x/\mathrm{d}u$:

$$\int \frac{1}{x^{3/2}(x-1)^{1/2}}\,\mathrm{d}x = \int \frac{(1-u)^2}{u^{1/2}} \cdot \frac{1}{(1-u)^2}\,\mathrm{d}u = \int u^{-1/2}\,\mathrm{d}u$$
$$= 2u^{1/2} + c = 2\sqrt{\frac{x-1}{x}} + c.$$

Part (ii)

In part (i), $u = \frac{x-1}{x}$ was the ratio of the two linear factors. Here we copy that idea and take

$$u = \frac{x-2}{x+1}, \qquad \text{equivalently} \qquad x+1 = \frac{3}{1-u}, \quad x-2 = \frac{3u}{1-u},$$

since $x-2 = (x+1)-3$. For $x > 2$ we get $0 < u < 1$, and $\dfrac{\mathrm{d}x}{\mathrm{d}u} = \dfrac{3}{(1-u)^2}$. Then

$$(x-2)^{3/2}(x+1)^{1/2} = \frac{3^{3/2}u^{3/2}}{(1-u)^{3/2}} \cdot \frac{3^{1/2}}{(1-u)^{1/2}} = \frac{9u^{3/2}}{(1-u)^2},$$

and so

$$\begin{aligned}\int \frac{1}{(x-2)^{3/2}(x+1)^{1/2}}\,\mathrm{d}x &= \int \frac{(1-u)^2}{9u^{3/2}} \cdot \frac{3}{(1-u)^2}\,\mathrm{d}u = \frac{1}{3}\int u^{-3/2}\,\mathrm{d}u \\ &= -\frac{2}{3}u^{-1/2} + c = -\frac{2}{3}\sqrt{\frac{x+1}{x-2}} + c.\end{aligned}$$

Differentiating confirms it. With $f = \frac{x+1}{x-2}$ we have $f' = -\frac{3}{(x-2)^2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(-\tfrac{2}{3}\sqrt{f}\right) = -\frac{f'}{3\sqrt{f}} = \frac{1}{(x-2)^2}\sqrt{\frac{x-2}{x+1}} = \frac{1}{(x-2)^{3/2}(x+1)^{1/2}}.$$

Part (iii)

This time no substitution is given. The integrand is not a pure power of a ratio because of the extra factor $x-1$, but the two square roots are of the same type as before.

Ask yourself Both earlier answers were square roots of a ratio of the linear factors. What happens if we make such a square root the new variable?

Try

$$t = \sqrt{\frac{x-2}{3x-2}}.$$

As x increases from 2 to ∞, $\frac{x-2}{3x-2} = \frac{1}{3} - \frac{4}{3(3x-2)}$ increases from 0 towards $\frac{1}{3}$, so t increases from 0 to $\frac{1}{\sqrt{3}}$ and $0 \le t < \frac{1}{\sqrt{3}}$ throughout. Solving $t^2(3x-2) = x-2$ for x gives

$$x = \frac{2(1-t^2)}{1-3t^2},$$

where $1-3t^2 > 0$. From this,

$$x-1 = \frac{1+t^2}{1-3t^2}, \qquad 3x-2 = \frac{4}{1-3t^2},$$

and

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{2[-2t(1-3t^2)+6t(1-t^2)]}{(1-3t^2)^2} = \frac{8t}{(1-3t^2)^2}.$$

The two square roots combine neatly, since $\sqrt{x-2} = t\sqrt{3x-2}$:

$$(x-2)^{1/2}(3x-2)^{1/2} = t\,(3x-2) = \frac{4t}{1-3t^2}.$$

So the whole denominator is

$$(x-1)(x-2)^{1/2}(3x-2)^{1/2} = \frac{4t(1+t^2)}{(1-3t^2)^2},$$

and once again everything cancels against $\mathrm{d}x/\mathrm{d}t$:

$$\int_2^\infty \frac{\mathrm{d}x}{(x-1)(x-2)^{1/2}(3x-2)^{1/2}} = \int_0^{1/\sqrt{3}} \frac{(1-3t^2)^2}{4t(1+t^2)} \cdot \frac{8t}{(1-3t^2)^2}\,\mathrm{d}t = \int_0^{1/\sqrt{3}} \frac{2}{1+t^2}\,\mathrm{d}t$$
$$= 2\Big[\arctan t\Big]_0^{1/\sqrt{3}} = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3},$$

as required.

The original integral is improper at both ends, but it converges. Near $x = 2$ the integrand behaves like $\frac{1}{2}(x-2)^{-1/2}$, and for large x it behaves like $\frac{1}{\sqrt{3}}x^{-2}$. The substitution turns it into an ordinary integral of a continuous function over a finite interval, so no limits need to be taken separately.

Checks

- Differentiate the answers to (i) and (ii). For (i), $\frac{\mathrm{d}}{\mathrm{d}x} 2\sqrt{1-\frac{1}{x}} = \frac{1}{x^2}\left(1-\frac{1}{x}\right)^{-1/2} = x^{-3/2}(x-1)^{-1/2}$.
- Signs: the integrand in (ii) is positive, so the antiderivative must increase. It does, since $\sqrt{\frac{x+1}{x-2}}$ decreases towards 1 as x grows.
- Numerical: $\int_{1.5}^{3}$ in (i) and $\int_{2.5}^{4}$ in (ii), evaluated by Simpson's rule, agree with the antiderivatives (0.47829… and 0.70974…). The integral in (iii), computed directly by splitting at $x = 3$, gives $1.0471975\ldots = \frac{\pi}{3}$.

Common slips

- Forgetting to change the limits in (iii). The upper limit is $t = \frac{1}{\sqrt{3}}$, not ∞, which is where the $\frac{\pi}{6}$ comes from.
- Writing the answer to (i) or (ii) in terms of u and not converting back to x, as the question asks.
- Mishandling the half-powers of $1-u$. Collect them carefully: $(1-u)^{-3/2}(1-u)^{-1/2} = (1-u)^{-2}$.

Beyond STEP

Integrands built from $\sqrt{(x-a)(x-b)}$ can always be made rational by a substitution like ours. This is part of the classical theory of integrating rational functions of x and $\sqrt{\text{quadratic}}$ (Euler's substitutions). Part (iii) is also an improper integral over an infinite range, and making sense of these with limits is a standard first-year analysis topic. Abbott's *Understanding Analysis* treats improper integrals carefully, and Siklos's *Advanced Problems in Mathematics* has more STEP integrals of this kind.

WORKED SOLUTION

STEP II 2020, Question 2

Pure • Separable Differential Equations

STEP II 2020, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The differential equation separates, and after integrating, the natural thing to look at is the product xy, which turns out to be a power of 2. The identity $(x+y)^2-(x-y)^2=4xy$ then rewrites everything in terms of the sum $s=x+y$ and the difference $d=x-y$. In those variables each curve becomes "d^2 equals a function of s", and the sketches come from asking where that function is non-negative.

Solution

Part (i)

Factorise the right-hand side:

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{y(kx-1)}{x(1-ky)} \quad\Longrightarrow\quad \frac{1-ky}{y}\,\mathrm{d}y=\frac{kx-1}{x}\,\mathrm{d}x.$$

Integrating, $\ln|y|-ky=kx-\ln|x|+c$, that is

$$\ln|xy|=k(x+y)+c.$$

On both curves x and y have the same sign, so $|xy|=xy$. Since $k=\ln 2$, exponentiating gives $xy=A\,2^{x+y}$ for a positive constant A.

Curve C_1. At $(1,1)$ we get $1=4A$, so $A=\frac{1}{4}$ and $4xy=2^{x+y}$. Now $4xy=(x+y)^2-(x-y)^2$, so

$$(x-y)^2=(x+y)^2-2^{x+y},$$

with $x,y>0$. This is the required result.

Curve C_2. At $(-1,-1)$ we get $1=A\cdot 2^{-2}$, so $A=4$ and $4xy=16\cdot 2^{x+y}=2^{x+y+4}$. Hence

$$(x-y)^2=(x+y)^2-2^{x+y+4},$$

with $x,y<0$.

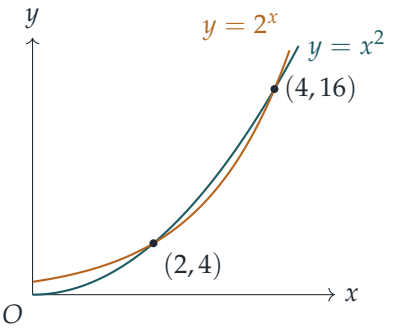
Symmetry. Swapping x and y leaves $x+y$ unchanged and leaves $(x-y)^2 = (y-x)^2$ unchanged, so each equation is unchanged. The sign conditions ($x, y > 0$, or $x, y < 0$) are unchanged too. So whenever (a, b) lies on a curve, so does (b, a), its reflection in $y = x$. Hence $y = x$ is a line of symmetry of each curve.

Part (ii)

For $x \geq 0$, the curves $y = x^2$ and $y = 2^x$ meet at $(2, 4)$ and $(4, 16)$. These are the only meeting points. To see this, compare logarithms: let $g(x) = 2\ln x - x\ln 2$ for $x > 0$. Then $g'(x) = \frac{2}{x} - \ln 2$, so g increases up to $x = 2/\ln 2$ and decreases after it. So g has at most two zeros, and $g(2) = g(4) = 0$. Hence

$$x^2 \geq 2^x \iff 2 \leq x \leq 4 \qquad (x > 0),$$

and 2^x lies above x^2 on $[0, 2)$ and on $(4, \infty)$.



Now take a point of C_1 and put $s = x + y > 0$. The left-hand side of the equation of C_1 is a square, so

$$s^2 - 2^s = (x-y)^2 \geq 0, \qquad \text{that is} \qquad s^2 \geq 2^s.$$

By the sketch, this forces $2 \leq s \leq 4$. So C_1 lies between the lines $x + y = 2$ and $x + y = 4$.

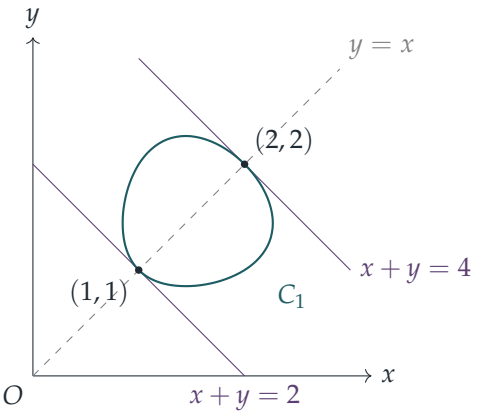
Ask yourself Is every value of s in $[2, 4]$ actually reached, and what does the curve look like at the ends?

For each $s \in [2, 4]$, put $d = \pm\sqrt{s^2 - 2^s}$ and $x = \frac{1}{2}(s + d)$, $y = \frac{1}{2}(s - d)$. Since $2^s > 0$, we have $|d| < s$, so x and y are both positive. So every such s gives a point of C_1, two points when $2 < s < 4$. The curve is therefore a closed loop:

- At $s = 2$ and $s = 4$ we have $d = 0$, giving the points $(1, 1)$ and $(2, 2)$ on $y = x$. Both points satisfy $kxy - y = -(x - kxy)$, so the gradient there is -1. The loop touches the lines $x + y = 2$ and $x + y = 4$ at these points.
- The gradient is 0 where $x = 1/\ln 2 \approx 1.44$ and infinite where $y = 1/\ln 2$. At $x = 1/\ln 2$ the loop has $y \approx 0.85$ (its lowest point) and $y \approx 2.27$ (its highest).

- Its greatest width across $y = x$ is where $s^2 - 2^s$ is largest, at $s \approx 3.21$, where $|d| \approx 1.02$.

The loop is an oval, symmetric about $y = x$, squeezed between the two lines:



Part (iii)

Now $s = x + y < 0$. Write $t = -s > 0$. As before, we need

$$s^2 \geq 2^{s+4} \iff t^2 \geq 2^{4-t} \iff t^2\, 2^t \geq 16.$$

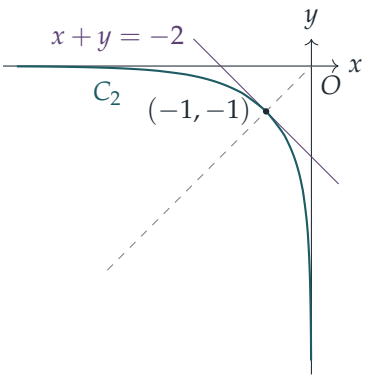
The function $t^2\, 2^t$ is increasing for $t > 0$ and equals 16 at $t = 2$. So the condition is $t \geq 2$, that is $x + y \leq -2$. (Equivalently, on a sketch of $y = x^2$ and $y = 2^{x+4}$ for $x < 0$, the parabola is above only when $x \leq -2$.) For every $s \leq -2$ the same construction as in (ii) gives two points with $x, y < 0$, except at $s = -2$, which gives the single point $(-1, -1)$. So C_2 is unbounded this time:

- It meets $y = x$ only at $(-1, -1)$, where the gradient is again -1, so it touches the line $x + y = -2$ there.
- For $x, y < 0$ we have $kx - 1 < 0$ and $1 - ky > 0$, so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y(kx - 1)}{x(1 - ky)} < 0,$$

  since the numerator is positive and the denominator negative. So y decreases steadily as x increases.
- From $xy = 4 \cdot 2^{x+y}$: as $x \to -\infty$, the product xy tends to 0, so $y \to 0^-$ (in fact $y \approx 4 \cdot 2^x / x$). By symmetry, $y \to -\infty$ as $x \to 0^-$. The negative axes are asymptotes.

So C_2 looks like one branch of a rectangular hyperbola in the third quadrant, but it hugs the axes much more tightly:



Checks

- Differentiate $4xy = 2^{x+y}$ implicitly: $4y + 4xy' = k\,2^{x+y}(1+y') = 4kxy(1+y')$. Rearranging gives $y' = \dfrac{kxy - y}{x - kxy}$, the original equation. The same works for C_2.
- $(2,2)$ lies on C_1: $4 \cdot 2 \cdot 2 = 16 = 2^4$. On C_2, the point with $x + y = -4$ is about $(-3.936, -0.064)$, and $xy \approx 0.252$ while $4 \cdot 2^{-4} = 0.25$. The extreme points of the oval quoted above were confirmed by computer.

Common slips

- Forgetting the modulus signs when integrating $1/x$ and $1/y$. For C_2 they matter, and the constant comes out as $A = 4$, not $\frac{1}{4}$.
- Showing only $(x+y)^2 \geq 2^{x+y}$ and stopping. You must also explain why this confines $x + y$ to $[2, 4]$, which is where the sketch of x^2 against 2^x comes in.
- Drawing C_2 crossing the axes. On C_2 both coordinates stay negative, and the axes are asymptotes.

Beyond STEP

Rotating coordinates to $s = x + y$, $d = x - y$ turns a symmetric equation into "$d^2 = f(s)$". The curve then exists exactly where $f(s) \geq 0$: it is a closed loop when that set is a bounded interval and unbounded when it is a half-line. The same reasoning draws phase portraits from a conserved quantity: $\ln(xy) - k(x+y)$ is constant along every solution, just as energy is in the Lotka–Volterra predator–prey equations, whose orbits are closed loops for the same reason. Boyce and DiPrima, *Elementary Differential Equations*, covers both ideas.

WORKED SOLUTION

STEP II 2020, Question 3

Pure • Log-Concave Sequences

STEP II 2020, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

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The key idea

Property L says that each ratio u_{r+1}/u_r is at most the one before it, so once the sequence stops rising it can never start rising again. That is exactly unimodality. In part (ii) the recurrence has a repeated root α in its auxiliary equation, and the combination $u_r - \alpha u_{r-1}$ turns it into a simple geometric sequence. Everything else follows from that one substitution.

Solution

Part (i)

Suppose the sequence has property L and $u_{r-1} \geq u_r$, where $2 \leq r \leq n-1$. Since $u_{r-1} > 0$, we can divide property L by it:

$$u_{r+1} \leq \frac{u_r^2}{u_{r-1}} = u_r \cdot \frac{u_r}{u_{r-1}} \leq u_r,$$

where the last step uses $u_r/u_{r-1} \leq 1$ and $u_r > 0$. So $u_{r-1} \geq u_r \implies u_r \geq u_{r+1}$.

Now take any sequence with property L. If $u_1 \leq u_2 \leq \cdots \leq u_n$, it is unimodal with $k = n$. Otherwise there is a first index k with $u_k > u_{k+1}$ (so $k \leq n-1$). By the choice of k,

$$u_1 \leq u_2 \leq \cdots \leq u_k.$$

Since $u_k \geq u_{k+1}$, the implication just proved (with $r = k+1$) gives $u_{k+1} \geq u_{k+2}$; applying it again with $r = k+2$ gives $u_{k+2} \geq u_{k+3}$, and so on by induction up to $u_{n-1} \geq u_n$. Hence $u_k \geq u_{k+1} \geq \cdots \geq u_n$, and the sequence is unimodal.

Part (ii)

Write $v_r = u_r - \alpha u_{r-1}$ for $2 \leq r \leq n$. For $3 \leq r \leq n$, the recurrence gives

$$v_r = 2\alpha u_{r-1} - \alpha^2 u_{r-2} - \alpha u_{r-1} = \alpha\left(u_{r-1} - \alpha u_{r-2}\right) = \alpha v_{r-1}.$$

So (v_r) is geometric with ratio α, and $v_r = \alpha^{r-2}v_2$, that is,

$$u_r - \alpha u_{r-1} = \alpha^{r-2}(u_2 - \alpha u_1) \qquad (2 \le r \le n).$$

(For $r = 2$ this is trivially true, and each step up in r multiplies both sides by α.)

For $2 \le r \le n-1$ we have $3 \le r+1 \le n$, so we may substitute $u_{r+1} = 2\alpha u_r - \alpha^2 u_{r-1}$:

$$u_r^2 - u_{r-1}u_{r+1} = u_r^2 - 2\alpha u_r u_{r-1} + \alpha^2 u_{r-1}^2 = (u_r - \alpha u_{r-1})^2.$$

Positive and unimodal. Suppose $u_2 > \alpha u_1 > 0$. Since $\alpha > 0$, this gives $u_1 > 0$, and $u_2 > 0$ too. For the later terms, the first identity gives

$$u_r = \alpha u_{r-1} + \alpha^{r-2}(u_2 - \alpha u_1) > \alpha u_{r-1},$$

because $\alpha^{r-2} > 0$ and $u_2 - \alpha u_1 > 0$. So if $u_{r-1} > 0$ then $u_r > 0$, and by induction every term is positive. The second identity now gives $u_{r-1}u_{r+1} = u_r^2 - (u_r - \alpha u_{r-1})^2 \le u_r^2$ for $2 \le r \le n-1$. That is property L for a sequence of positive terms, so the sequence is unimodal by part (i).

The case $u_1 = 1$, $u_2 = 2$. Let $w_r = (2-r)\alpha^{r-1} + 2(r-1)\alpha^{r-2}$. Then $w_1 = 1 = u_1$ and $w_2 = 2 = u_2$. With these starting values the first identity reads

$$u_r = \alpha u_{r-1} + (2-\alpha)\alpha^{r-2}.$$

Suppose $u_{r-1} = w_{r-1}$ for some r with $3 \le r \le n$. Then

$$\begin{aligned} u_r &= \alpha\left[(3-r)\alpha^{r-2} + 2(r-2)\alpha^{r-3}\right] + 2\alpha^{r-2} - \alpha^{r-1} \\ &= (2-r)\alpha^{r-1} + 2(r-1)\alpha^{r-2} = w_r, \end{aligned}$$

so by induction $u_r = (2-r)\alpha^{r-1} + 2(r-1)\alpha^{r-2}$ for $1 \le r \le n$. (We could equally run a two-step induction using the original recurrence; the first-order form is quicker.)

The maximum when $\alpha = 1 - \frac{1}{N}$. Here $\frac{1}{2} \le \alpha < 1$, so $u_2 = 2 > \alpha = \alpha u_1 > 0$, and by the work above the sequence is positive and unimodal. To find the peak, look at consecutive differences. From $u_{r+1} = \alpha u_r + (2-\alpha)\alpha^{r-1}$ and $\alpha - 1 = -\frac{1}{N}$,

$$u_{r+1} - u_r = (2-\alpha)\alpha^{r-1} - \frac{u_r}{N} = \frac{\alpha^{r-2}}{N}\Big[N\alpha(2-\alpha) - (2-r)\alpha - 2(r-1)\Big].$$

Now $N\alpha = N - 1$ and $2 - \alpha = \frac{N+1}{N}$, so $N\alpha(2-\alpha) = N - \frac{1}{N}$, and $(2-r)\alpha = 2 - r - \frac{2-r}{N}$. The bracket simplifies to

$$N - \frac{1}{N} - 2 + r + \frac{2-r}{N} - 2r + 2 = (N-r) - \frac{r-1}{N} = \frac{N^2 + 1 - r(N+1)}{N},$$

so

$$u_{r+1} - u_r = \frac{\alpha^{r-2}}{N^2}\Big(N^2 + 1 - r(N+1)\Big).$$

Ask yourself Where exactly does $N^2+1-r(N+1)$ change sign, and could it ever be zero?

It is zero only at $r = \dfrac{N^2+1}{N+1} = N-1+\dfrac{2}{N+1}$. Since $N \geq 2$, we have $0 < \frac{2}{N+1} < 1$, so this value lies strictly between the integers $N-1$ and N. As $\alpha^{r-2} > 0$:

- for $r \leq N-1$, $u_{r+1} > u_r$;
- for $r \geq N$, $u_{r+1} < u_r$.

Hence $u_1 < u_2 < \cdots < u_N$ and $u_N > u_{N+1} > \cdots > u_n$. Since $N \leq n$, the term u_N exists, and u_r is largest when $r = N$ (strictly, with no ties).

Checks

Take $N = 5$, so $\alpha = \frac{4}{5}$. The recurrence gives 1, 2, 2.56, 2.816, 2.8672, 2.78528, 2.62144, …, rising to its peak at $r = 5$ and falling afterwards, as claimed. The difference formula predicts $u_6 - u_5 = \alpha^3(26-30)/25 = -0.08192$, which matches. Also $u_3 = 2\alpha \cdot 2 - \alpha^2 = 2.56$ agrees with the closed form $-\alpha^2 + 4\alpha$. The identities and the location of the peak were also confirmed by computer, in exact arithmetic, for many values of α, u_1, u_2 and for N up to 29.

Common slips

- In part (i), dividing by u_{r-1} without noting that it is positive. The argument fails for sequences with negative terms.
- Forgetting to show the terms are positive in part (ii) before appealing to part (i), which needs positivity.
- Showing only that $u_{N+1} < u_N$ and $u_{N-1} < u_N$. Unimodality makes this enough, but you then need to say so; it is cleaner to find the sign of every difference.

Beyond STEP

Property L is called *log-concavity*: $\ln u_r$ is a concave function of r. Log-concave sequences turn up all over combinatorics and probability. The binomial coefficients $\binom{n}{r}$ are an example, and Newton's inequalities say that the coefficients of a polynomial with only real roots are log-concave. The recurrence in part (ii) is a linear recurrence with a repeated root, which is why the solution has the form $(A+Br)\alpha^r$, just as $(A+Bx)\mathrm{e}^{\alpha x}$ arises for repeated roots in differential equations. For the differential-equation side of this parallel, see Boyce and DiPrima, *Elementary Differential Equations*; for problems in this spirit, Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2020, Question 4

Pure • Triangle Inequality

STEP II 2020, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Everything rests on the triangle inequality: each side is strictly shorter than the sum of the other two. In fact only one inequality ever needs checking, the one for the *longest* side, because the other two hold automatically. For the converse, build the triangle with compasses: put down one side and swing two circles from its ends.

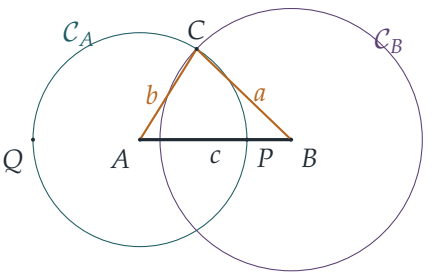
Solution

Part (i)

Label the vertices A, B, C with $BC = a$, $CA = b$, $AB = c$. The straight segment AB is the shortest route from A to B, so the route $A \to C \to B$, of length $b + a$, is at least c. It can't equal c: that happens only when C lies on the segment AB, and then there is no triangle. Hence $c < a + b$. The same argument for the other two pairs of vertices gives $a < b + c$ and $b < a + c$.

Part (ii)

Draw a segment AB of length c. Draw the circle $\mathcal{C}_A$ with centre A and radius b, and the circle $\mathcal{C}_B$ with centre B and radius a. If they meet at a point C off the line AB, then ABC is a triangle with $CA = b$, $CB = a$ and $AB = c$.



Why must they meet? Let P and Q be the points where $\mathcal{C}_A$ crosses the line AB, with P on the same side of A as B. Then

$$PB = |c - b|, \qquad QB = c + b.$$

Now $|c - b| < a$, because $c - b < a$ and $b - c < a$ are two of the given inequalities; and $c + b > a$ is the third. So as a point travels round the upper half of $\mathcal{C}_A$ from P to Q, its distance from B changes continuously from a value below a to a value above a, and therefore equals a somewhere. That point lies on both circles, and it is not on the line AB, since the only points of $\mathcal{C}_A$ on that line are P and Q, whose distances from B are not a. This is the required point C, so a triangle with sides a, b, c exists.

Part (iii)

Parts (i) and (ii) together say that three positive numbers are the sides of a triangle exactly when each is less than the sum of the other two. Write ℓ for the largest of the three. If ℓ is less than the sum of the other two, the other two inequalities follow at once, since each of the smaller numbers is at most ℓ and the remaining number is positive. So we only ever need to test the largest length.

(A) Always. All three lengths are positive. Using $c < a + b$,

$$c + 1 < a + b + 1 < (a + 1) + (b + 1),$$

and the other two inequalities follow in the same way.

(B) Sometimes but not always. For an equilateral triangle the three lengths are $1, 1, 1$, which do form a triangle. But take $a = b = 1$ and $c = \frac{1}{10}$, which is a genuine triangle since $1 < 1 + \frac{1}{10}$. The lengths become 1, 10, $\frac{1}{10}$, and $10 > 1 + \frac{1}{10}$, so no triangle is possible.

(C) Never. The three lengths are symmetric in a, b, c, so we may label so that $a \geq b \geq c$. Then

$$|c - a| = a - c = (a - b) + (b - c) = |a - b| + |b - c|.$$

The largest length equals the sum of the other two, so the strict inequality fails. (If two of a, b, c are equal, one of the lengths is 0, which fails too.)

(D) Always. The set $\{a^2 + bc,\ b^2 + ca,\ c^2 + ab\}$ doesn't change when we swap any two of a, b, c. (Swapping a and b, for instance, gives $b^2 + ac$, $a^2 + cb$, $c^2 + ba$.) So it is enough to prove one of the three inequalities, say $a^2 + bc < (b^2 + ca) + (c^2 + ab)$, for every triangle. The difference is

$$\begin{aligned}(b^2 + ca) + (c^2 + ab) - (a^2 + bc) &= (b^2 - bc + c^2) + a(b + c) - a^2 \\ &= \left(b - \tfrac{1}{2}c\right)^2 + \tfrac{3}{4}c^2 + a(b + c - a).\end{aligned}$$

The first bracket is positive because $c > 0$, and $a(b+c-a) > 0$ because $a > 0$ and $a < b+c$. So the difference is positive. The other two inequalities follow by relabelling, and all three lengths are clearly positive, so they always form a triangle.

Part (iv)

The roles of a, b, c are interchangeable, so we may label so that $a \geq b \geq c$. All three of $f(a)$, $f(b)$, $f(c)$ are positive. Because f is increasing, $f(a) \geq f(b) \geq f(c)$, so by the remark at the start of part (iii) we only need

$$f(a) < f(b) + f(c).$$

Ask yourself How can the condition on $f(x)/x$ help compare $f(a)$ with $f(b) + f(c)$?

It lets us bound $f(b)$ and $f(c)$ *below* by multiples of $f(a)$. Since $f(x)/x$ decreases as x increases, and $a \geq b$,

$$\frac{f(b)}{b} \geq \frac{f(a)}{a}, \qquad \text{so} \qquad f(b) \geq \frac{b}{a} f(a),$$

with equality allowed only when $b = a$. In the same way $f(c) \geq \frac{c}{a} f(a)$. Adding,

$$f(b) + f(c) \;\geq\; \frac{b+c}{a} f(a) \;>\; f(a),$$

where the last step uses $b + c > a$ (from the triangle) and $f(a) > 0$. So the largest of the three new lengths is less than the sum of the other two, and $f(a)$, $f(b)$, $f(c)$ are the sides of a triangle.

Checks

- For (D), try $a = 2$, $b = 3$, $c = 4$: the lengths are 16, 17, 22, and $22 < 16 + 17$. The identity, relabelled for the longest length $c^2 + ab$, predicts the gap $(a^2 - ab + b^2) + c(a + b - c) = (4 - 6 + 9) + 4 \cdot 1 = 11$, and indeed $16 + 17 - 22 = 11$.
- For (C), the triangle $3, 4, 5$ gives $1, 1, 2$: flat, exactly as predicted.
- A computer check on 200 000 random triangles confirmed (A), (D) and (C) (always, always, never), found (B) holding for about 40% of them, and confirmed part (iv) for $f(x) = \sqrt{x}$, $\ln(1 + x)$, $x/(1 + x)$, $x^{0.3}$ and $1 - \mathrm{e}^{-x}$.

Common slips

- Answering (B) with "never" or "always" after testing one triangle: a "sometimes" answer needs one example of each kind.
- In (D), checking only one inequality without explaining why the other two follow: the symmetry (or the longest-side remark) is what justifies it.

- In (iv), writing $f(a)/a < f(b)/b$ when a might equal b; the inequality is then $\leq$, and the strictness has to come from $b + c > a$.

Beyond STEP

The triangle inequality is one of the three axioms of a *metric*, the abstract notion of distance at the heart of first-year analysis and topology. Part (iv) is a classic fact there: if d is a metric and f is increasing with $f(0) = 0$ and $f(x)/x$ decreasing, then $f \circ d$ is a metric too; this is why $d/(1 + d)$ is a standard way to make a bounded metric. Bartle and Sherbert, *Introduction to Real Analysis*, is a good place to meet these ideas.

WORKED SOLUTION

STEP II 2020, Question 5

Pure • Digit Sums and Divisibility

STEP II 2020, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Every power of 10 leaves remainder 1 on division by 9, so a number and its digit sum always agree modulo 9. That single fact turns each equation into a strong divisibility condition. A size argument then finishes the job: the digit sum grows only like $9n$ while the number itself grows like 10^{n-1}, so only a handful of candidates are left to test.

Solution

Part (i)

With $x = \sum_{r=0}^{n-1} a_r 10^r$ and $\mathrm{d}(x) = \sum_{r=0}^{n-1} a_r$,

$$x - \mathrm{d}(x) = \sum_{r=0}^{n-1} a_r \left(10^r - 1\right).$$

Each term has $a_r \geq 0$ and $10^r - 1 \geq 0$, so the sum is non-negative. Also $10^r - 1 = 99\ldots9$ (r nines) $= 9 \times 11\ldots1$, which is divisible by 9 (for $r = 0$ it is 0). So every term, and hence $x - \mathrm{d}(x)$, is divisible by 9. ■

In the language of congruences: $x \equiv \mathrm{d}(x) \pmod 9$ for every positive integer x. We use this repeatedly below.

Part (ii)

The equivalence. Since $\mathrm{d}(x) \equiv x \pmod 9$,

$$x - 44\,\mathrm{d}(x) \equiv x - 44x = -43x \equiv 2x \pmod 9,$$

because $-43 = -45 + 2$. So $x - 44\,\mathrm{d}(x)$ is a multiple of 9 exactly when $9 \mid 2x$. As $\gcd(2, 9) = 1$, $9 \mid 2x$ if and only if $9 \mid x$. This proves both directions at once. ■

The bounds. Suppose $x = 44\,\mathrm{d}(x)$ and x has n digits. Each digit is at most 9, so $\mathrm{d}(x) \leq 9n$ and $x = 44\,\mathrm{d}(x) \leq 396n$. And an n-digit number has leading digit $a_{n-1} \geq 1$,

so $x \geq 10^{n-1}$. Together,

$$10^{n-1} \leq x \leq 396n.$$

For $n = 5$ this would need $10\,000 \leq 1980$, which is false. Moreover, if $10^{n-1} > 396n$ for some $n \geq 5$ then $10^n = 10 \cdot 10^{n-1} > 3960n \geq 396(n+1)$, so by induction the inequality $10^{n-1} \leq 396n$ fails for every $n \geq 5$. Hence $n \leq 4$. ■

Ask yourself How can we cut the search down from all numbers up to 9999 to just a few?

Use the equivalence we have just proved. If $x = 44\,\mathrm{d}(x)$, then $x - 44\,\mathrm{d}(x) = 0$, which is a multiple of 9, so $9 \mid x$. Then $9 \mid 44\,\mathrm{d}(x)$, which forces $9 \mid \mathrm{d}(x)$ since $\gcd(44, 9) = 1$. Therefore $x = 44\,\mathrm{d}(x)$ is a multiple of $44 \times 9 = 396$.

Finding the solution. Since $n \leq 4$, we have $x \leq 396 \times 4 = 1584$, so x is one of 396, 792, 1188, 1584. Their digit sums are all 18, so $44\,\mathrm{d}(x) = 792$ in every case. The only one that works is

$$x = 792, \qquad \mathrm{d}(792) = 18, \quad 44 \times 18 = 792.$$

Every solution lies in our list of four candidates, and only 792 works, so there are no further values. ■

Part (iii)

Write $y = \mathrm{d}(x)$ and $z = \mathrm{d}(y)$, so that the equation is $x = 107z$, with $z = x/107 > 0$.

Divisibility. By part (i), $x \equiv y \equiv z \pmod 9$. So

$$0 = x - 107z \equiv z - 107z = -106z \pmod 9.$$

Now $106 = 99 + 7$, so $9 \mid 7z$, and since $\gcd(7, 9) = 1$ we get $9 \mid z$.

Size. Part (i) also gives $\mathrm{d}(m) \leq m$ for any positive integer m. So if x has n digits then $z = \mathrm{d}(y) \leq y \leq 9n$, and

$$10^{n-1} \leq x = 107z \leq 963n.$$

For $n = 5$ this needs $10\,000 \leq 4815$, which is false. The same induction as in part (ii) (now $10^n > 9630n \geq 963(n+1)$) shows that it fails for all $n \geq 5$, so $n \leq 4$.

Ask yourself The bound $z \leq 36$ still leaves several multiples of 9. Can we do better?

Yes, by applying the digit sum a second time. With $n \leq 4$, $y = \mathrm{d}(x) \leq 36$. Among $1, 2, \ldots, 36$ the largest digit sum is $\mathrm{d}(29) = 11$, because a two-digit number below 37 has tens digit at most 3, and tens digit 3 allows units digit at most 6 (digit sum at most 9), whereas tens digit 2 allows $2 + 9 = 11$. So $1 \leq z \leq 11$, and z is a multiple of 9. Hence $z = 9$ and

$$x = 107 \times 9 = 963.$$

Check: $\mathrm{d}(963) = 18$, $\mathrm{d}(18) = 9$ and $107 \times 9 = 963$. So $x = 963$ satisfies the equation. Any solution must have $z = 9$, and hence $x = 963$, so there are no further values. ■

Checks

- Spot-check part (i): $249 - 15 = 234 = 9 \times 26$.
- A computer search over every $x < 10^6$ finds 792 as the only solution of $x = 44\,\mathrm{d}(x)$ and 963 as the only solution of $x = 107\,\mathrm{d}(\mathrm{d}(x))$, in line with the bounds above.
- Notice $792 = 44 \times 18$ and $963 = 107 \times 9$: both are multiples of 9, as the divisibility arguments require.

Common slips

- Concluding only that x is a multiple of 9 in part (ii). The stronger fact that $\mathrm{d}(x)$ is also a multiple of 9, so that $396 \mid x$, is what makes the search short.
- Stating $n \leq 4$ after checking only $n = 5$. You need a reason why 10^{n-1} stays ahead of $396n$ for all larger n; the induction (or the observation that the left side grows by a factor of 10 while the right grows by a factor of at most $\frac{6}{5}$) supplies it.
- In part (iii), forgetting that $z > 0$, which is what rules out $z = 0$.

Beyond STEP

The fact that $x \equiv \mathrm{d}(x) \pmod 9$ is the theory behind "casting out nines", an old way of checking arithmetic. It is the first example of working in the ring $\mathbb{Z}/9\mathbb{Z}$, where 10 behaves like 1; the same idea with $10 \equiv -1$ gives the alternating-sum test for divisibility by 11. Equations mixing a number with functions of its digits are typically solved exactly as here: congruences cut down the possibilities and growth estimates make the list finite. Davenport's *The Higher Arithmetic* is a lovely introduction to congruences.

WORKED SOLUTION

STEP II 2020, Question 6

Pure • Matrix Powers and Trace

STEP II 2020, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.4, *Matrices That Return to the Start*: https://beyondstep.keithmcnulty.org/ch07/finite-order.html

The key idea

Every real 2×2 matrix satisfies its own characteristic equation,

$$\mathbf{M}^2 - t\,\mathbf{M} + \delta\,\mathbf{I} = \mathbf{0}, \qquad t = \operatorname{tr}(\mathbf{M}),\ \delta = \det(\mathbf{M}),$$

the Cayley–Hamilton theorem (it is a two-line check by multiplying out). Once $\mathbf{M}^2$ is written in terms of $\mathbf{M}$ and $\mathbf{I}$, any equation such as $\mathbf{M}^2 = \mathbf{I}$ becomes "$t\mathbf{M}$ is a multiple of $\mathbf{I}$", which forces either $t = 0$ or $\mathbf{M}$ itself to be a multiple of $\mathbf{I}$. Parts (iii) and (iv) then come from applying part (ii) to $\mathbf{M}^2$ and $\mathbf{M}^4$ instead of $\mathbf{M}$.

Throughout, write $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $t = a + d$ and $\delta = ad - bc$.

Solution

Part (i)

Multiplying out,

$$\mathbf{M}^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & bc + d^2 \end{pmatrix} = (a+d)\begin{pmatrix} a & b \\ c & d \end{pmatrix} - (ad - bc)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

since, for example, $(a+d)a - (ad-bc) = a^2 + bc$ and $(a+d)b = ab + bd$. So

$$\mathbf{M}^2 = t\mathbf{M} - \delta\mathbf{I}. \qquad (*)$$

Taking the trace of $(*)$, and using $\operatorname{tr}(\mathbf{I}) = 2$,

$$\operatorname{tr}(\mathbf{M}^2) = t \cdot t - 2\delta = \operatorname{tr}(\mathbf{M})^2 - 2\det(\mathbf{M}),$$

as required. (Directly: $a^2 + 2bc + d^2 = (a+d)^2 - 2(ad - bc)$.)

Part (ii)

First equivalence, (⇒). Suppose $\mathbf{M}^2 = \mathbf{I}$ and $\mathbf{M} \neq \pm\mathbf{I}$. Substituting into $(*)$ gives

$$t\,\mathbf{M} = (1+\delta)\,\mathbf{I}.$$

If $t \neq 0$ we could divide to get $\mathbf{M} = k\mathbf{I}$ with $k = (1+\delta)/t$ real; then $\mathbf{M}^2 = k^2\mathbf{I} = \mathbf{I}$ gives $k = \pm 1$, so $\mathbf{M} = \pm\mathbf{I}$, which is excluded. Hence $t = 0$, and then $(1+\delta)\mathbf{I} = \mathbf{0}$, so $\delta = -1$.

(⇐). If $t = 0$ and $\delta = -1$, then $(*)$ gives $\mathbf{M}^2 = 0 \cdot \mathbf{M} + \mathbf{I} = \mathbf{I}$. Also $\operatorname{tr}(\pm\mathbf{I}) = \pm 2 \neq 0$, so $\mathbf{M} \neq \pm\mathbf{I}$.

Second equivalence, (⇒). If $\mathbf{M}^2 = -\mathbf{I}$, then $(*)$ gives $t\mathbf{M} = (\delta - 1)\mathbf{I}$. If $t \neq 0$ then $\mathbf{M} = k\mathbf{I}$ for a real k, and $\mathbf{M}^2 = k^2\mathbf{I} = -\mathbf{I}$ would need $k^2 = -1$, impossible for real k. So $t = 0$, and then $\delta = 1$.

(⇐). If $t = 0$ and $\delta = 1$, then $(*)$ gives $\mathbf{M}^2 = -\mathbf{I}$ at once.

> **Ask yourself** Where exactly did we use that $\mathbf{M}$ is *real*?

Only in the second equivalence: $k^2 = -1$ has no real solution. For complex matrices, $\mathbf{M} = \mathrm{i}\mathbf{I}$ satisfies $\mathbf{M}^2 = -\mathbf{I}$ with trace $2\mathrm{i} \neq 0$, so the result would be false.

Part (iii)

(⇐) is immediate: if $\mathbf{M}^2 = \pm\mathbf{I}$ then $\mathbf{M}^4 = (\pm\mathbf{I})^2 = \mathbf{I}$.

(⇒). Let $\mathbf{N} = \mathbf{M}^2$, a real 2×2 matrix with $\mathbf{N}^2 = \mathbf{I}$. Either $\mathbf{N} = \pm\mathbf{I}$, which is what we want, or $\mathbf{N} \neq \pm\mathbf{I}$, in which case part (ii) says $\det(\mathbf{N}) = -1$. But

$$\det(\mathbf{N}) = \det(\mathbf{M})^2 \geq 0,$$

so the second case cannot happen. Hence $\mathbf{M}^2 = \pm\mathbf{I}$.

The condition for $\mathbf{M}^4 = -\mathbf{I}$. Again put $\mathbf{N} = \mathbf{M}^2$. By the second equivalence of part (ii) applied to $\mathbf{N}$,

$$\mathbf{M}^4 = -\mathbf{I} \iff \operatorname{tr}(\mathbf{N}) = 0 \text{ and } \det(\mathbf{N}) = 1.$$

Using part (i) and $\det(\mathbf{N}) = \delta^2$, this says

$$t^2 - 2\delta = 0 \quad \text{and} \quad \delta^2 = 1.$$

Now $\delta^2 = 1$ gives $\delta = \pm 1$, and $t^2 = 2\delta \geq 0$ rules out $\delta = -1$. So $\delta = 1$ and $t^2 = 2$. Every step was an equivalence, so

$$\boxed{\mathbf{M}^4 = -\mathbf{I} \iff \det(\mathbf{M}) = 1 \text{ and } \operatorname{tr}(\mathbf{M}) = \pm\sqrt{2}}.$$

Part (iv)

Applying part (iii) to $\mathbf{M}^2$: $\mathbf{M}^8 = \mathbf{I} \iff \mathbf{M}^4 = \pm\mathbf{I}$, so the matrices with $\mathbf{M}^8 = \mathbf{I}$ are exactly those with

$$\mathbf{M}^2 = \pm\mathbf{I} \qquad \text{or} \qquad \det(\mathbf{M}) = 1,\ \operatorname{tr}(\mathbf{M}) = \pm\sqrt{2}.$$

Rotations and reflections are the *orthogonal* matrices: their columns are perpendicular unit vectors (rotations have determinant 1, reflections determinant -1). So we just need a matrix in this list whose columns are not unit vectors. For example,

$$\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

has trace 0 and determinant -1, so $\mathbf{M}^2 = \mathbf{I}$ by part (ii) (check: $\mathbf{M}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$), and hence $\mathbf{M}^8 = \mathbf{I}$. Its second column $(1, -1)$ has length $\sqrt{2} \neq 1$, so it is neither a rotation nor a reflection. (It is an "oblique reflection": it fixes every vector along the x-axis and reverses every vector along $(1, -2)$, and these two directions are not perpendicular.)

An example from the more interesting family is $\mathbf{M} = \begin{pmatrix} \sqrt{2} & 1 \\ -1 & 0 \end{pmatrix}$: determinant 1 and trace $\sqrt{2}$, so $\mathbf{M}^4 = -\mathbf{I}$ and $\mathbf{M}^8 = \mathbf{I}$, yet its first column has length $\sqrt{3}$. It is a "distorted" rotation through 45°: it has order exactly 8, like the rotation, but does not preserve lengths.

Checks

- The rotation through 45°, $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, has determinant 1 and trace $\sqrt{2}$, and four 45° turns make a half-turn, which is $-\mathbf{I}$: consistent with part (iii).
- Squaring $\mathbf{M} = \begin{pmatrix} \sqrt{2} & 1 \\ -1 & 0 \end{pmatrix}$ gives $\begin{pmatrix} 1 & \sqrt{2} \\ -\sqrt{2} & -1 \end{pmatrix}$, which has trace 0 and determinant 1, so its square is $-\mathbf{I}$ by part (ii).
- A computer search over all matrices with entries in $\{-3, -\frac{5}{2}, \ldots, 3\}$ confirmed both equivalences in part (ii) and the first in part (iii) exactly.

Common slips

- Dividing by t without first handling $t = 0$; the whole argument in part (ii) is a case split on whether t is zero.
- In part (iii), forgetting that $\det(\mathbf{M}^2) = \det(\mathbf{M})^2$ cannot be negative; that is precisely what rules out the "other" square roots of $\mathbf{I}$.
- Giving a rotation through 45° as the example in part (iv): it satisfies $\mathbf{M}^8 = \mathbf{I}$ but is exactly what is excluded.

Beyond STEP

The identity $(*)$ is the 2×2 case of the Cayley–Hamilton theorem: every square matrix satisfies its characteristic polynomial. Questions like "which $\mathbf{M}$ satisfy $\mathbf{M}^n = \mathbf{I}$?" are

really about eigenvalues: here they must be eighth roots of unity, and a real matrix of order 8 is *similar* to a rotation without being one. Strang's *Introduction to Linear Algebra* develops eigenvalues, similarity and the characteristic polynomial in this spirit.

WORKED SOLUTION

STEP II 2020, Question 7

Pure • Complex Transformations

STEP II 2020, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The map $z \mapsto w = \dfrac{2}{z-2}$ is a shift followed by an inversion, and inversion sends lines that miss the origin to circles through the origin. To *prove* that w lies on a particular circle, the cleanest route is to guess the centre c and then show that $|w-c|$ simplifies to a constant. Writing $w-c$ as a single fraction, the numerator and denominator turn out to be conjugates (or nearly so), which is what makes the modulus constant.

Solution

Throughout we use $|\alpha/\beta| = |\alpha|/|\beta|$ and $|x+yi| = |x-yi| = |y+xi| = \sqrt{x^2+y^2}$ for real x, y.

Part (i)

With $z = 3+ti$ we have $z-2 = 1+ti$, so

$$w - 1 = \frac{2}{1+ti} - 1 = \frac{2-(1+ti)}{1+ti} = \frac{1-ti}{1+ti}.$$

The numerator and denominator are complex conjugates, so they have the same modulus, $\sqrt{1+t^2}$, which is never zero. Hence

$$|w-1| = \frac{\sqrt{1+t^2}}{\sqrt{1+t^2}} = 1,$$

which does not depend on t. Every z on the line $\mathrm{Re}(z) = 3$ has the form $3+ti$ for some real t, so for every such z the point w is at distance 1 from 1: it lies on the circle with centre 1 and radius 1.

The line V. Now $z = p+ti$, and it helps to write $a = p-2$, which is a non-zero real number because $p \neq 2$. Then $z-2 = a+ti$, which is never zero. When $p = 3$ we had

centre $1 = 1/a$, so we try the centre $1/a$ in general:

$$w - \frac{1}{a} = \frac{2}{a+ti} - \frac{1}{a} = \frac{2a-(a+ti)}{a(a+ti)} = \frac{a-ti}{a(a+ti)}.$$

Again $|a - ti| = |a + ti|$, so

$$\left|w - \frac{1}{p-2}\right| = \frac{1}{|a|} = \frac{1}{|p-2|}.$$

So w lies on the circle with

$$\text{centre } \frac{1}{p-2}, \qquad \text{radius } \frac{1}{|p-2|}.$$

The modulus sign in the radius matters: if $p < 2$ the centre is on the negative real axis, but a radius must be positive.

Ask yourself How would we have found the centre $1/(p-2)$ if we hadn't been told the $p = 3$ case?

The point $z = p$ (where $t = 0$) gives $w = 2/a$, and as $t \to \pm\infty$ we get $w \to 0$. These two points are at opposite ends of a diameter (the circle is symmetric about the real axis, and they are its two real points), so the centre is their midpoint, $1/a$.

For the imaginary part, multiply top and bottom by the conjugate:

$$w = \frac{2(a-ti)}{a^2+t^2}, \qquad \text{so} \qquad \mathrm{Im}(w) = \frac{-2t}{a^2+t^2}.$$

The denominator is positive, so $\mathrm{Im}(w) > 0$ exactly when $t < 0$. Hence the points of V with $\mathrm{Im}(w) > 0$ are

$$z = p + ti \text{ with } t < 0, \quad \text{that is, the points of } V \text{ below the real axis } (\mathrm{Im}(z) < 0).$$

Part (ii)

Now $z = s + qi$ with s real, and we write $u = s - 2$, so $z - 2 = u + qi$, which is non-zero since $q \neq 0$. Here the two 'ends' of the circle are $w = 0$ (as $u \to \pm\infty$) and $w = 2/(qi) = -2i/q$ (at $u = 0$), suggesting the centre $-i/q$. Check:

$$w + \frac{i}{q} = \frac{2}{u+qi} + \frac{i}{q} = \frac{2q+i(u+qi)}{q(u+qi)} = \frac{q+ui}{q(u+qi)}.$$

Now $|q + ui| = \sqrt{q^2+u^2} = |u + qi|$, so

$$\left|w + \frac{i}{q}\right| = \frac{1}{|q|}.$$

So w lies on the circle with

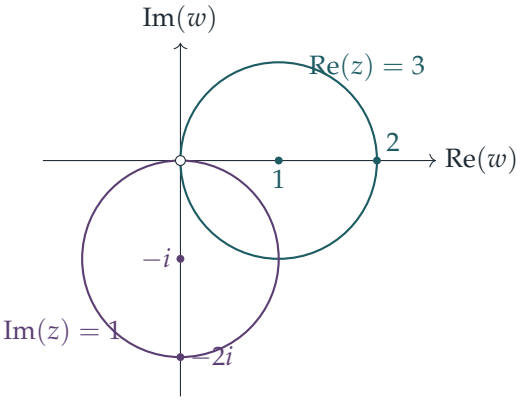
$$\text{centre } -\frac{i}{q}, \qquad \text{radius } \frac{1}{|q|}.$$

For the real part,

$$w = \frac{2(u - qi)}{u^2 + q^2}, \qquad \text{so} \qquad \mathrm{Re}(w) = \frac{2u}{u^2 + q^2},$$

with positive denominator. So $\mathrm{Re}(w) > 0$ exactly when $u > 0$, that is, for

$$z = s + qi \text{ with } s > 2, \quad \text{the points of } H \text{ to the right of } \mathrm{Re}(z) = 2.$$



The figure shows the image of $\mathrm{Re}(z) = 3$ (from part (i)) and of $\mathrm{Im}(z) = 1$ (part (ii) with $q = 1$). Both circles pass through the origin, but $w = 0$ itself is never reached, since $2/(z - 2) \neq 0$; it corresponds to z 'at infinity' along the line.

Checks

- Take $p = 3$ in the formula for V: centre $1/(3 - 2) = 1$ and radius 1, matching the first result.
- Spot values: $z = 3 - i$ gives $w = 2/(1 - i) = 1 + i$, which has $|w - 1| = 1$ and $\mathrm{Im}(w) = 1 > 0$, consistent with $t = -1 < 0$. For H with $q = 1$, $z = 3 + i$ gives $w = 1 - i$, which is at distance 1 from $-i$ and has positive real part, as $3 > 2$.
- The radius $1/|p - 2|$ grows as $p \to 2$: the line V then passes close to the point $z = 2$ that is sent to infinity, so its image should be huge. A computer check over thousands of random p, q and points on each line confirmed both circles and both sign conditions.

Common slips

- Giving the radius as $1/(p - 2)$ or $1/q$, which is negative when $p < 2$ or $q < 0$.
- Getting the sign of $\mathrm{Im}(w)$ the wrong way round: dividing by $z - 2$ reflects in the real axis, so $\mathrm{Im}(w)$ has the opposite sign to $\mathrm{Im}(z)$.

Beyond STEP

The map $z \mapsto 2/(z-2)$ is a Möbius transformation, $z \mapsto (az+b)/(cz+d)$. These maps always send 'generalised circles' (circles and lines) to generalised circles, and they are the basic geometric tool of a first course in complex analysis, from conformal mapping to hyperbolic geometry. Needham's *Visual Complex Analysis* has a beautiful chapter on them, with pictures that make results like this one obvious.

WORKED SOLUTION

STEP II 2020, Question 8

Pure • Polynomials and Integration

STEP II 2020, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Because $\mathrm{F}' = \mathrm{f}$, every root of f is a turning point of F, and each area condition becomes a statement about the value of F at one of those turning points. Once we know F has double roots at 0 and c, the sum $\mathrm{F}(x) + \mathrm{F}(c - x)$ collapses to a single sign-carrying factor, $c - 2h$. Comparing that sign with what we know about the largest and smallest values of F on $[0, c]$ forces the symmetry.

Solution

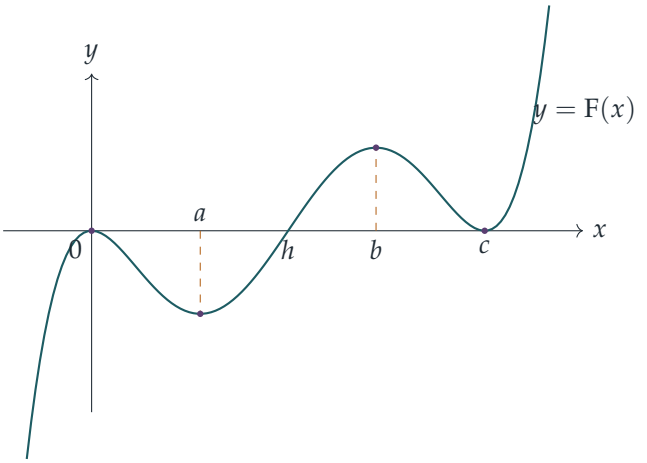
Since f is monic with roots $0, a, b, c$, we have $\mathrm{f}(x) = x(x-a)(x-b)(x-c)$. So $\mathrm{f} < 0$ on $(0, a)$, $\mathrm{f} > 0$ on (a, b), $\mathrm{f} < 0$ on (b, c), and $\mathrm{f} > 0$ for $x < 0$ and $x > c$. Write A for the area between 0 and a. The conditions say that the area between b and c is A and the area between a and b is $2A$, with $A > 0$.

Part (i)

Because $\mathrm{F}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t$, the integral over an interval where $\mathrm{f} < 0$ is minus the area there. So

$$\mathrm{F}(0) = 0, \qquad \mathrm{F}(a) = -A, \qquad \mathrm{F}(b) = -A + 2A = A, \qquad \mathrm{F}(c) = A - A = 0.$$

Since $\mathrm{F}' = \mathrm{f}$, the turning points of F are at $x = 0, a, b, c$. The sign of f changes from $+$ to $-$ at 0 and at b, giving maxima there, and from $-$ to $+$ at a and at c, giving minima. F is a quintic with positive leading coefficient $\frac{1}{5}$, so $\mathrm{F}(x) \to \pm\infty$ as $x \to \pm\infty$.



The curve has a maximum at $(0,0)$, a minimum at $x=a$ with $y=-A$, a maximum at $x=b$ with $y=A$, and a minimum touching the axis at $(c,0)$.

The form of F. F is a polynomial of degree 5 with leading coefficient $\frac{1}{5}$. At $x=0$ we have $\mathrm{F}(0)=0$ and $\mathrm{F}'(0)=\mathrm{f}(0)=0$, so $x=0$ is a root of multiplicity at least 2. In the same way $\mathrm{F}(c)=0=\mathrm{F}'(c)$, so c is a root of multiplicity at least 2. That accounts for four of the five linear factors:

$$\mathrm{F}(x)=\tfrac{1}{5}x^2(x-c)^2(x-h)$$

for some real h. Also, $\mathrm{F}(a)=-A<0<A=\mathrm{F}(b)$, so F has a root strictly between a and b. Neither 0 nor c lies in (a,b), so this root must be h. Hence $a<h<b$, and in particular $0<h<c$.

The sum. Note that $(c-x)^2\big((c-x)-c\big)^2=x^2(x-c)^2$. So

$$\begin{aligned}\mathrm{F}(x)+\mathrm{F}(c-x)&=\tfrac{1}{5}x^2(x-c)^2\big[(x-h)+(c-x-h)\big]\\&=\tfrac{1}{5}x^2(x-c)^2(c-2h).\end{aligned}$$

Part (ii)

On $[0,c]$, F is strictly decreasing on $[0,a]$, strictly increasing on $[a,b]$ and strictly decreasing on $[b,c]$. So on $[0,c]$ the smallest value is $\mathrm{F}(a)=-A$: the only other candidate is the endpoint value $\mathrm{F}(c)=0>-A$. This minimum is attained only at $x=a$. So for $0\le x\le c$,

$$\mathrm{F}(b)+\mathrm{F}(x)\ \ge\ \mathrm{F}(b)+\mathrm{F}(a)=A-A=0,$$

with strict inequality when $x\neq a$.

Ask yourself Which value of x makes $\mathrm{F}(b)+\mathrm{F}(x)$ look like the sum from part (i)?

Take $x = c - b$, which lies in $[0, c]$ because $0 < b < c$. By part (i),

$$\mathrm{F}(b) + \mathrm{F}(c-b) = \tfrac{1}{5}b^2(b-c)^2(c-2h).$$

If $c - b \neq a$, the left-hand side is strictly positive. Since $\frac{1}{5}b^2(b-c)^2 > 0$, it follows that $c - 2h > 0$. So either $c - b = a$ or $c > 2h$, as required.

Now run the same argument the other way up. On $[0, c]$ the largest value of F is $\mathrm{F}(b) = A$, beating the endpoint values $\mathrm{F}(0) = \mathrm{F}(c) = 0$, and it is attained only at $x = b$. So for $0 \leq x \leq c$,

$$\mathrm{F}(a) + \mathrm{F}(x) \leq \mathrm{F}(a) + \mathrm{F}(b) = 0,$$

with strict inequality when $x \neq b$. Taking $x = c - a \in [0, c]$ gives $\frac{1}{5}a^2(c-a)^2(c-2h) \leq 0$, strictly if $c - a \neq b$. So either $c - a = b$ or $c < 2h$.

If $c \neq a + b$, both alternatives give $c > 2h$ and $c < 2h$, which is impossible. Hence $c = a + b$. Then $c - b = a$, so

$$0 = \mathrm{F}(b) + \mathrm{F}(a) = \mathrm{F}(b) + \mathrm{F}(c-b) = \tfrac{1}{5}b^2a^2(c-2h).$$

Because $a, b \neq 0$, this gives $c = 2h$.

Part (iii)

With $h = \frac{1}{2}c$, we have $\mathrm{F}(x) = \frac{1}{5}u\left(x - \frac{1}{2}c\right)$, where $u = x^2(x-c)^2$, so that $u' = 2x(x-c)(2x-c)$. Differentiating with the product rule,

$$\begin{aligned}
\mathrm{f}(x) = \mathrm{F}'(x) &= \tfrac{1}{5}\left[2x(x-c)(2x-c)\left(x - \tfrac{1}{2}c\right) + x^2(x-c)^2\right] \\
&= \tfrac{1}{5}x(x-c)\left[(2x-c)^2 + x(x-c)\right] \\
&= \tfrac{1}{5}x(x-c)\left(5x^2 - 5cx + c^2\right) \\
&= x(x-c)\left(x^2 - cx + \tfrac{1}{5}c^2\right).
\end{aligned}$$

(So $a, b = \frac{1}{2}c \mp \frac{c}{2\sqrt{5}}$, which do add up to c.)

> **Ask yourself** The roots are symmetric about $x = \frac{1}{2}c$. Can we use that to make the second derivative easy?

Yes. Put $y = x - \frac{1}{2}c$. Then $x(x-c) = y^2 - \frac{1}{4}c^2$ and $x^2 - cx + \frac{1}{5}c^2 = y^2 - \frac{1}{20}c^2$, so

$$\mathrm{f} = \left(y^2 - \tfrac{1}{4}c^2\right)\left(y^2 - \tfrac{1}{20}c^2\right) = y^4 - \tfrac{3}{10}c^2y^2 + \tfrac{1}{80}c^4.$$

Since $\mathrm{d}y/\mathrm{d}x = 1$,

$$\mathrm{f}''(x) = 12y^2 - \tfrac{3}{5}c^2,$$

which is zero exactly when $y^2 = \frac{1}{20}c^2$. At those points the factor $y^2 - \frac{1}{20}c^2$ of f vanishes, so $\mathrm{f} = 0$. Each is a simple root of the quadratic f'' in y, so f'' changes sign there, and they are genuine points of inflection. So both points of inflection lie on the x-axis. In fact they are the points $(a, 0)$ and $(b, 0)$.

Checks

- Take $c = 2$, so that $a = 1 - \frac{1}{\sqrt{5}} \approx 0.5528$ and $b \approx 1.4472$. Numerical integration of f gives areas 0.05724, 0.11449 and 0.05724 over $[0, a]$, $[a, b]$ and $[b, c]$: the outer two are equal, and each is half the middle one.
- Expanding $x(x-a)(x-b)(x-c)$ with $a + b = c$ and $ab = \frac{1}{5}c^2$ reproduces the expression in part (iii).
- With $c = 2$, a finite-difference estimate of f'' at $x = a \approx 0.5528$ is zero to within rounding, and it changes sign between $x = 0.54$ and $x = 0.56$.

Common slips

- Forgetting that $\mathrm{F}'(0) = \mathrm{F}'(c) = 0$ as well as $\mathrm{F}(0) = \mathrm{F}(c) = 0$, and so missing the double roots.
- Not justifying $0 < h < c$: the sign change of F between a and b is what places h.
- Using only the first inequality in part (ii). It gives "$c = a + b$ or $c > 2h$", which alone is not enough. The reversed inequality with $\mathrm{F}(a)$ is needed to close the argument.

Beyond STEP

The argument in part (ii) uses the extreme value theorem: a continuous function on a closed interval attains a greatest and a least value, and here we also know exactly where. The quartic's symmetry about $x = \frac{1}{2}c$, which the area conditions force, is an example of how integral conditions can pin down the shape of a polynomial. Abbott's *Understanding Analysis* gives a careful treatment of the extreme value theorem and of the fundamental theorem of calculus, which links F and f.

WORKED SOLUTION

STEP II 2020, Question 9

Mechanics • Colliding Projectiles

STEP II 2020, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

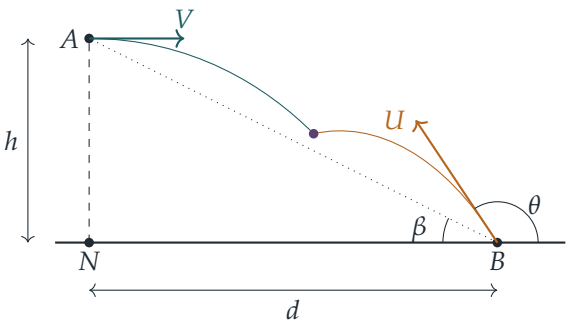
www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Put both particles in one coordinate system and write down where each is at time t. Both feel the same downward acceleration g, so the $-\frac{1}{2}gt^2$ terms cancel when we compare heights: relative to each other, the particles move in straight lines at constant velocity. That makes the collision time very simple, and every later part is a short consequence of it.

Solution

Model both particles as point masses moving freely under gravity, with no air resistance. Take N as origin, the x-axis along NB (so $B = (d, 0)$, $A = (0, h)$) and the y-axis vertically upwards. The particle from A moves towards B and the particle from B moves towards N, so their horizontal velocities are opposite; this is the set-up the given result describes.



At time t after projection, the particle from A is at

$$x_A = Vt, \qquad y_A = h - \tfrac{1}{2}gt^2,$$

and the particle from B is at

$$x_B = d - Ut\cos\theta, \qquad y_B = Ut\sin\theta - \tfrac{1}{2}gt^2.$$

They collide at time T if both coordinates agree. The heights agree when

$$h - \tfrac{1}{2}gT^2 = UT\sin\theta - \tfrac{1}{2}gT^2 \quad \Longrightarrow \quad T = \frac{h}{U\sin\theta},$$

which is the only possible collision time ($\sin\theta > 0$ as θ is acute). The horizontal positions must agree at this same time:

$$VT = d - UT\cos\theta \quad \Longrightarrow \quad (V + U\cos\theta)\,\frac{h}{U\sin\theta} = d.$$

Multiplying by $U\sin\theta$ gives $Vh + Uh\cos\theta = Ud\sin\theta$, and dividing by U,

$$d\sin\theta - h\cos\theta = \frac{Vh}{U},$$

as required.

Part (i)

From the right-angled triangle ANB, $\tan\beta = h/d$, with $0 < \beta < \frac{1}{2}\pi$. Since V, h, U are positive, the result just shown gives $d\sin\theta - h\cos\theta > 0$. Dividing by $d\cos\theta > 0$ (as θ is acute),

$$\tan\theta > \frac{h}{d} = \tan\beta.$$

Both angles lie in $(0, \frac{1}{2}\pi)$, where tan is strictly increasing, so $\theta > \beta$.

Part (ii)

> **Ask yourself** Nothing in the algebra so far has used the ground. What physical condition have we not yet imposed?

The collision must happen before either particle reaches the ground, so the common height at time T must be at least 0:

$$h - \tfrac{1}{2}gT^2 \geq 0 \quad \Longleftrightarrow \quad h \geq \frac{gh^2}{2U^2\sin^2\theta} \quad \Longleftrightarrow \quad U^2\sin^2\theta \geq \frac{gh}{2}.$$

(We divided by $h > 0$ and multiplied by $U^2\sin^2\theta > 0$.) Since $U\sin\theta > 0$, taking positive square roots gives

$$U\sin\theta \geq \sqrt{\frac{gh}{2}}.$$

Equality corresponds to the particles meeting just as they reach the ground.

Part (iii)

Write $r = AB = \sqrt{d^2 + h^2}$, so that $d = r\cos\beta$ and $h = r\sin\beta$. Then

$$d\sin\theta - h\cos\theta = r(\sin\theta\cos\beta - \cos\theta\sin\beta) = r\sin(\theta - \beta),$$

and the collision condition becomes

$$r\sin(\theta-\beta)=\frac{Vr\sin\beta}{U}\quad\Longrightarrow\quad\frac{U}{V}=\frac{\sin\beta}{\sin(\theta-\beta)}.$$

By part (i), $\theta-\beta>0$; also $\theta-\beta<\theta<\frac{1}{2}\pi$. So $0<\sin(\theta-\beta)<1$, and since $\sin\beta>0$,

$$\frac{U}{V}=\frac{\sin\beta}{\sin(\theta-\beta)}>\sin\beta.$$

The final part

The height of the collision is $h-\frac{1}{2}gT^2$, so

$$\text{collision above } \tfrac{1}{2}h \iff h-\tfrac{1}{2}gT^2>\tfrac{1}{2}h \iff gT^2<h.$$

The particle from B has vertical velocity $U\sin\theta-gt$, so

$$\text{it is rising at time } T \iff U\sin\theta-gT>0 \iff gT<U\sin\theta.$$

Ask yourself How can we compare $gT<U\sin\theta$ with $gT^2<h$?

Use the collision time itself: $T=h/(U\sin\theta)$ means $U\sin\theta=h/T$. So, multiplying by $T>0$,

$$gT<U\sin\theta \iff gT<\frac{h}{T} \iff gT^2<h.$$

Both conditions are equivalent to $gT^2<h$, so the particles collide above height $\frac{1}{2}h$ if and only if the particle from B is still moving upwards at the collision.

Checks

- Take $g=9.8$, $h=10$, $d=20$, $U=20$, $\theta=\frac{1}{4}\pi$. The collision condition gives $V=U(d\sin\theta-h\cos\theta)/h=10\sqrt{2}\approx14.14$, and $T=1/\sqrt{2}\approx0.707$ s. Then both particles are at $x=10$, $y=10-4.9\times\frac{1}{2}=7.55$, above $\frac{1}{2}h=5$; the particle from B has vertical velocity $10\sqrt{2}-9.8/\sqrt{2}\approx7.21>0$, rising, as the final part predicts. Also $U/V=\sqrt{2}>\sin\beta=1/\sqrt{5}$.
- Gravity-free intuition: with $g=0$ the particles move in straight lines, and the condition $U/V=\sin\beta/\sin(\theta-\beta)$ is just the sine rule in the triangle formed by A, B and the meeting point. Gravity shifts both particles down by the same amount, so the condition is unchanged.
- A computer check over thousands of random choices of h, d, U, θ confirmed the collision, all three inequalities, and the equivalence in the final part.

Common slips

- Forgetting the ground in part (ii): the inequality comes from the collision happening at a height of at least zero, not from anything in the collision equation.

- Concluding $\theta > \beta$ from $\tan\theta > \tan\beta$ without saying that both angles are acute, where tan is increasing.
- In part (iii), stopping at $\sin(\theta - \beta) \leq 1$; the strict inequality needs $\theta - \beta < \frac{1}{2}\pi$.

Beyond STEP

The cancellation of the $-\frac{1}{2}gt^2$ terms is an instance of changing to a freely falling frame of reference: in a frame accelerating with g, gravity disappears and both particles move in straight lines. This is the simplest case of the Galilean and accelerating-frame ideas that open a first university mechanics course, and later the starting point of Einstein's equivalence principle. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames carefully.

WORKED SOLUTION

STEP II 2020, Question 10

Mechanics • Energy and Equilibrium

STEP II 2020, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

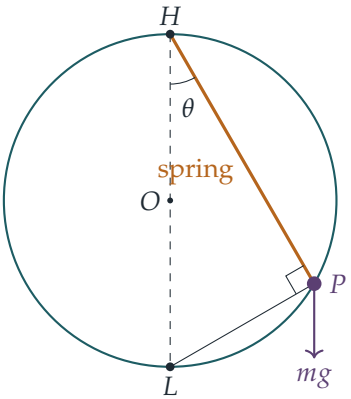
www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Because HL is a diameter, the angle in a semicircle gives the spring length and the height of P directly in terms of θ: $HP = 2a\cos\theta$, and P is $2a\cos^2\theta$ below H. The wire is smooth, so its reaction does no work. That makes the whole problem about one function, the potential energy as a function of $c = \cos\theta$. It turns out to be a quadratic in c, and completing the square gives both parts at once.

Solution

Model P as a particle on a smooth wire, and the spring as light and obeying Hooke's law (in tension or compression). Let O be the centre of the circle.



Geometry. Since HL is a diameter, $\angle HPL = 90°$, so in the right-angled triangle HPL with hypotenuse $2a$,

$$HP = 2a\cos\theta.$$

The vertical drop from H to P is $HP\cos\theta = 2a\cos^2\theta$. As P goes round from L towards H on either side, θ runs over $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\cos\theta > 0$.

Potential energy. Take H as the level of zero gravitational potential energy and write $c = \cos\theta$. The extension of the spring is $2ac - l$, so

$$\begin{aligned} V &= -2mga\,c^2 + \frac{\lambda}{2l}(2ac - l)^2 \\ &= \frac{2a(a\lambda - mgl)}{l}\,c^2 - 2a\lambda\,c + \frac{\lambda l}{2}. \end{aligned}$$

Part (i)

The reaction of the smooth wire does no work, so P is in equilibrium exactly where $\mathrm{d}V/\mathrm{d}\theta = 0$. (Equivalently, the forces along the tangent balance.) By the chain rule,

$$\frac{\mathrm{d}V}{\mathrm{d}\theta} = -\sin\theta\,\frac{\mathrm{d}V}{\mathrm{d}c} = -\sin\theta\left(\frac{4a(a\lambda - mgl)}{l}\,c - 2a\lambda\right).$$

At $\theta = \alpha$ with $0 < \alpha < \frac{\pi}{2}$ we have $\sin\alpha \neq 0$, so the bracket must vanish:

$$\frac{4a(a\lambda - mgl)}{l}\cos\alpha = 2a\lambda.$$

The right-hand side is non-zero, so $a\lambda \neq mgl$, and we can divide:

$$\cos\alpha = \frac{\lambda l}{2(a\lambda - mgl)},$$

as required.

For such an α to exist we need $0 < \cos\alpha < 1$.

- $\cos\alpha > 0$: the numerator λl is positive, so we need $a\lambda > mgl$.
- $\cos\alpha < 1$: multiplying up by the positive quantity $2(a\lambda - mgl)$ gives $\lambda l < 2a\lambda - 2mgl$, that is, $\lambda(2a - l) > 2mgl$.

Since $l < 2a$, the second condition is $\lambda > \dfrac{2mgl}{2a - l}$. This also implies the first, because $\dfrac{2mgl}{2a-l} > \dfrac{2mgl}{2a} = \dfrac{mgl}{a}$. So an equilibrium with $\alpha > 0$ exists only if $\lambda > \dfrac{2mgl}{2a-l}$. (Conversely, if this holds, the formula gives a genuine angle $\alpha \in (0, \frac{\pi}{2})$, and by symmetry $-\alpha$ is an equilibrium too.)

Part (ii)

> **?** **Ask yourself** Is there a neat way to write V that makes $\cos\alpha$ appear by itself?

Yes: complete the square. Write $K = \dfrac{2a(a\lambda - mgl)}{l}$. From part (i), $\dfrac{a\lambda - mgl}{l} = \dfrac{\lambda}{2\cos\alpha}$, so

$$K = \frac{a\lambda}{\cos\alpha}, \qquad \frac{2a\lambda}{K} = 2\cos\alpha.$$

Hence
$$V = K\left(c^2 - 2c\cos\alpha\right) + \frac{\lambda l}{2} = \frac{a\lambda}{\cos\alpha}(c - \cos\alpha)^2 + \text{constant}.$$
(As a check, this has its minimum at $c = \cos\alpha$, the equilibrium found in part (i).) We assume, as part (i) requires for the equilibrium to exist, that $a\lambda > mgl$, so $\cos\alpha > 0$ and $K > 0$.

At L, $\theta = 0$ and $c = 1$. Conservation of energy between L and a general point gives
$$\tfrac{1}{2}mv^2 = \tfrac{1}{2}mu^2 + \frac{a\lambda}{\cos\alpha}\Big[(1 - \cos\alpha)^2 - (c - \cos\alpha)^2\Big].$$
Multiplying by $\dfrac{\cos\alpha}{a\lambda} > 0$,
$$\frac{mv^2\cos\alpha}{2a\lambda} = D - (c - \cos\alpha)^2, \qquad \text{where } D = (1 - \cos\alpha)^2 + \frac{mu^2}{2a\lambda}\cos\alpha. \qquad (*)$$
So if P comes to rest at $\theta = \beta$, then $v = 0$ there and
$$(\cos\alpha - \cos\beta)^2 = (1 - \cos\alpha)^2 + \frac{mu^2}{2a\lambda}\cos\alpha,$$
which is the required equation.

Ask yourself The equation has two roots for $\cos\beta$. Which one is the actual stopping point, and when is it before H?

As P rises from L towards H, c decreases from 1 towards 0. By $(*)$, P keeps moving as long as $(c - \cos\alpha)^2 < D$, so it first stops where c meets the nearer end of the interval $(\cos\alpha - \sqrt{D},\ \cos\alpha + \sqrt{D})$ when moving down from 1. Now $D \geq (1 - \cos\alpha)^2$ (strictly if $u > 0$), so $\cos\alpha + \sqrt{D} > 1$ and $\cos\alpha - \sqrt{D} < 1$. Thus $c = 1$ lies inside the interval, and the stopping value is
$$\cos\beta = \cos\alpha - \sqrt{D}.$$
P stops before reaching H (where $c = 0$) exactly when this is positive, that is, when $\sqrt{D} < \cos\alpha$, or $D < \cos^2\alpha$:
$$\begin{aligned} & (1 - \cos\alpha)^2 + \frac{mu^2}{2a\lambda}\cos\alpha < \cos^2\alpha \\ \iff \quad & 1 - 2\cos\alpha + \frac{mu^2}{2a\lambda}\cos\alpha < 0 \\ \iff \quad & \frac{mu^2}{2a\lambda} < 2 - \sec\alpha, \end{aligned}$$
dividing by $\cos\alpha > 0$ in the last step. Hence P comes to rest before H only if
$$u^2 < \frac{2a\lambda}{m}(2 - \sec\alpha).$$
In particular this needs $\sec\alpha < 2$, i.e. $\alpha < \frac{\pi}{3}$. Otherwise P reaches H whatever its speed at L.

Checks

- Take $m = 1$, $g = 10$, $a = 1$, $l = 1$, $\lambda = 40$ (in consistent units). Then $\lambda > 2mgl/(2a - l) = 20$ and $\cos\alpha = 40/60 = \frac{2}{3}$. The critical value is $u^2 = 80(2 - \frac{3}{2}) = 40$. With $u^2 = 40$, $D = \frac{1}{9} + \frac{40}{80} \cdot \frac{2}{3} = \frac{4}{9}$, so $\cos\beta = \frac{2}{3} - \frac{2}{3} = 0$. The particle just reaches H, as it should.
- If $u = 0$ then $D = (1 - \cos\alpha)^2$ and $\cos\beta = 1$. A particle at rest at L stays there, which is right because L is also an equilibrium.
- A computer integration of the equation of motion, for many random parameter sets, stopped the particle at the predicted $\cos\beta$ exactly when u^2 was below the critical value.

Common slips

- Taking $HP = 2a\sin\theta$, or measuring the height from O with the wrong angle. Remember that the angle at the centre is 2θ, not θ.
- Only checking $\cos\alpha > 0$ in part (i). It is the condition $\cos\alpha < 1$ that gives the bound on λ.
- Choosing the root $\cos\alpha + \sqrt{D}$ for $\cos\beta$. It exceeds 1, so it isn't a position on the wire.

Beyond STEP

Writing the energy as $K(c - \cos\alpha)^2 + \text{constant}$ is the one-dimensional version of a key idea in undergraduate mechanics. For a system with one degree of freedom, the equilibria are the stationary points of the potential, their stability depends on whether those points are minima or maxima, and the regions of allowed motion are the places where the energy exceeds the potential. Here, when $\alpha > 0$ exists, L becomes a maximum of V and so an unstable equilibrium. The spring makes the bottom of the wire unstable! Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy diagrams and stability in this spirit.

WORKED SOLUTION

STEP II 2020, Question 11

Probability and Statistics • Runs in Coin Tossing

STEP II 2020, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The game has very little memory: all that matters at any moment is which face came up last and how long the current run is. In particular, as soon as the face changes, the game is in exactly the same position as if that new face had been the very first toss. So we condition on the first toss and write the probability that A wins, given a head first, in terms of the probability that A wins, given a tail first, and vice versa. That gives two simultaneous linear equations.

Solution

Throughout, $0 < p < 1$. (If $p = 0$ or $p = 1$ every toss is the same, the game ends on the second or third toss and the answers are trivial.) Write

$$\alpha = \mathrm{P}(\text{A wins} \mid \text{first toss H}), \qquad \beta = \mathrm{P}(\text{A wins} \mid \text{first toss T}).$$

Tosses are independent, so what happens after any toss depends only on the tosses still to come.

Part (i)

The game ends. The game is still going after n tosses only if no two consecutive tosses so far are the same, that is, the first n tosses alternate: $HTHT\ldots$ or $THTH\ldots$. For $n = 2m$ each of these two sequences contains m heads and m tails, so

$$\mathrm{P}(\text{not ended after } 2m \text{ tosses}) = 2p^m q^m = 2(pq)^m \leq 2\left(\tfrac{1}{4}\right)^m,$$

since $pq = p(1-p) = \frac{1}{4} - \left(p - \frac{1}{2}\right)^2 \leq \frac{1}{4}$. The event "the game never ends" is contained in every one of these events, so its probability is at most $2\left(\frac{1}{4}\right)^m$ for every m, and letting $m \to \infty$ shows that it is 0.

Given a head first. Look at the second toss.

- With probability p it is a head, and A wins at once.
- With probability q it is a tail. The head before it is now irrelevant: B needs one more tail, A needs two heads in a row, exactly as if this tail were the first toss. So A then wins with probability β.

Similarly, if the first toss is a tail, B wins at once if the second is a tail, and if the second is a head we are in the "head first" position. Hence

$$\alpha = p + q\beta, \qquad \beta = p\alpha.$$

Substituting the second equation into the first, $\alpha = p + pq\,\alpha$, so

$$\alpha = \frac{p}{1-pq},$$

as required ($1 - pq \geq \frac{3}{4} > 0$). Then $\beta = \dfrac{p^2}{1-pq}$.

Overall. Conditioning on the first toss,

$$\mathrm{P}(\text{A wins}) = p\alpha + q\beta = \frac{p^2 + p^2 q}{1-pq} = \frac{p^2(1+q)}{1-pq}.$$

Part (ii)

Now α and β refer to the game needing runs of three. Suppose the first toss is a head.

- HH then H, probability p^2: A wins.
- HHT, probability pq, or HT, probability q: the first tail since the start has just appeared and the earlier heads no longer count, so we are in the "tail first" position and A wins with probability β.

These cases cover everything, so

$$\alpha = p^2 + (q + pq)\,\beta,$$

which is the given result. Swapping the roles of the two faces (in the "tail first" position, TTT gives B the win and contributes 0),

$$\beta = (p + pq)\,\alpha.$$

Ask yourself How do the factors $q + pq$ and $p + pq$ turn into $1 - p^2$ and $1 - q^2$?

Since $p + q = 1$, we have $q + pq = q(1 + p) = (1 - p)(1 + p) = 1 - p^2$, and similarly $p + pq = p(1 + q) = 1 - q^2$. This also makes sense directly: $1 - p^2$ is the chance that the two tosses after an opening head are not both heads, which is exactly when the run is broken. So the equations are

$$\alpha = p^2 + (1 - p^2)\beta, \qquad \beta = (1 - q^2)\alpha,$$

and substituting gives $\alpha\left[1-(1-p^2)(1-q^2)\right]=p^2$, that is,

$$\alpha=\frac{p^2}{1-(1-p^2)(1-q^2)}.$$

(The denominator is positive, because $0<1-p^2<1$ and $0<1-q^2<1$.) Finally,

$$\mathrm{P}(\text{A wins})=p\alpha+q\beta=\alpha\big(p+q(1-q^2)\big)=\alpha\left(p+q-q^3\right)=\alpha\left(1-q^3\right),$$

so

$$\mathrm{P}(\text{A wins})=\frac{p^2(1-q^3)}{1-(1-p^2)(1-q^2)},$$

as required.

Part (iii)

The same argument works for runs of length a and b. Given a head first, either the next $a-1$ tosses are all heads (probability p^{a-1}) and A wins, or a tail appears among them (probability $1-p^{a-1}$). In the second case the game simply restarts from that tail, which puts us in the "tail first" position. Likewise, given a tail first, B wins with probability q^{b-1}, and otherwise a head turns up before B's run is complete. So

$$\alpha=p^{a-1}+\left(1-p^{a-1}\right)\beta,\qquad \beta=\left(1-q^{b-1}\right)\alpha,$$

and therefore

$$\alpha=\frac{p^{a-1}}{1-\left(1-p^{a-1}\right)\left(1-q^{b-1}\right)}.$$

Again the denominator is positive, since both brackets lie strictly between 0 and 1. Then

$$p\alpha+q\beta=\alpha\left(p+q-q^b\right)=\alpha\left(1-q^b\right),$$

so

$$\boxed{\mathrm{P}(\text{A wins})=\frac{p^{a-1}\left(1-q^b\right)}{1-\left(1-p^{a-1}\right)\left(1-q^{b-1}\right)}}$$

If you prefer, the denominator expands to $p^{a-1}+q^{b-1}-p^{a-1}q^{b-1}$.

The case $a=b=2$. The formula becomes

$$\frac{p(1-q^2)}{1-(1-p)(1-q)}=\frac{p(1-q)(1+q)}{1-pq}=\frac{p^2(1+q)}{1-pq},$$

using $1-p=q$, $1-q=p$ in the denominator and $1-q=p$ in the numerator. This agrees with part (i). Putting $a=b=3$ gives back part (ii) exactly.

Checks

- *Fair coin, part (i).* With $p=q=\frac{1}{2}$ the answer is $\frac{1}{4}\cdot\frac{3}{2}\big/\frac{3}{4}=\frac{1}{2}$, as symmetry demands. The same happens in (iii) whenever $a=b$ and $p=\frac{1}{2}$.

- *Extremes.* As $p \to 1$ the answer in (iii) tends to 1, and as $p \to 0$ it tends to 0, as it should.
- *Numbers.* With $p = 0.4$ the part (i) answer is $0.16 \times 1.6/0.76 \approx 0.3368$, and the part (ii) answer is ≈ 0.2713. A computer simulation of 200 000 games gave 0.336 and 0.271. For $p = 0.55$, $a = 4$, $b = 2$ the formula gives 0.2450, again matched by simulation.

Common slips

- Forgetting that once the face changes, the earlier run counts for nothing. That's exactly why the position after a change is the same as "that face first".
- In (ii), missing the case HHT: the run can be broken on the second *or* the third toss.
- Giving α as the final answer to "the probability that A wins". You still have to average over the first toss, $p\alpha + q\beta$.

Beyond STEP

This is a first-step analysis of an absorbing Markov chain. The states are "current face and length of run", and the absorption probabilities satisfy linear equations obtained by conditioning on the next step. The same technique gives expected waiting times for patterns in coin tossing, and leads on to Penney's game, where the order of the patterns produces some surprising non-transitive results. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains and absorption at the right level.

WORKED SOLUTION

STEP II 2020, Question 12

Probability and Statistics • Biased Dice Inequalities

STEP II 2020, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The probabilities must add up to 1, so the biases satisfy $\sum \varepsilon_i = 0$. Once that is written down, expanding $\sum(\frac{1}{n} + \varepsilon_i)^2$ makes every linear term vanish and leaves something that is visibly non-negative. Part (ii) is then part (i) in disguise: dividing positive numbers by their total turns them into the probabilities of some die.

Solution

Write $p_i = \mathrm{P}(X = i) = \frac{1}{n} + \varepsilon_i$. Since $\sum_{i=1}^{n} p_i = 1$,

$$1 = n \cdot \frac{1}{n} + \sum_{i=1}^{n} \varepsilon_i \quad \Longrightarrow \quad \sum_{i=1}^{n} \varepsilon_i = 0.$$

Also each $p_i \geq 0$, so $\varepsilon_i \geq -\frac{1}{n}$ for every i. All sums below run over $i = 1, \ldots, n$.

Part (i)

The two rolls are independent, so the probability that both show i is p_i^2, and these events are disjoint for different i. Hence

$$\begin{aligned} \mathrm{P}(\text{same score twice}) = \sum p_i^2 &= \sum \left(\frac{1}{n^2} + \frac{2\varepsilon_i}{n} + \varepsilon_i^2 \right) \\ &= \frac{1}{n} + \frac{2}{n} \sum \varepsilon_i + \sum \varepsilon_i^2 = \frac{1}{n} + \sum_{i=1}^{n} \varepsilon_i^2. \end{aligned}$$

For a fair die every $\varepsilon_i = 0$ and the probability is $\frac{1}{n}$. For a biased die not all the ε_i are zero, so $\sum \varepsilon_i^2 > 0$. A biased die is therefore *more* likely than a fair one to show the same score on two successive rolls.

Part (ii)

First rewrite the double sum. It adds $x_i x_j$ over all pairs with $j < i$, that is, over each unordered pair of distinct indices exactly once. Squaring $S = \sum x_i$ produces every x_i^2 once and every such pair twice, so

$$\sum_{i=2}^{n} \sum_{j=1}^{i-1} x_i x_j = \frac{1}{2}\left(S^2 - \sum x_i^2\right).$$

Ask yourself Where does a die come from, when the x_i are just any positive numbers?

Set $p_i = x_i / S$. These are positive and add up to 1, so they are the probabilities of some n-sided die. By part (i), $\sum p_i^2 = \frac{1}{n} + \sum \varepsilon_i^2 \geq \frac{1}{n}$ (with equality exactly when the die is fair, i.e. all x_i are equal). Multiplying by $S^2 > 0$,

$$\sum x_i^2 \geq \frac{S^2}{n}.$$

Therefore

$$\sum_{i=2}^{n} \sum_{j=1}^{i-1} x_i x_j = \frac{1}{2}\left(S^2 - \sum x_i^2\right) \leq \frac{1}{2}\left(S^2 - \frac{S^2}{n}\right) = \frac{n-1}{2n}\left(\sum_{i=1}^{n} x_i\right)^2,$$

which is the required inequality. Equality holds when all the x_i are equal.

Part (iii)

Now the probability is $\sum p_i^3$. Expanding, and using $\sum \varepsilon_i = 0$ again,

$$\begin{aligned}\sum p_i^3 &= \sum \left(\frac{1}{n^3} + \frac{3\varepsilon_i}{n^2} + \frac{3\varepsilon_i^2}{n} + \varepsilon_i^3\right) \\ &= \frac{1}{n^2} + \sum_{i=1}^{n}\left(\frac{3\varepsilon_i^2}{n} + \varepsilon_i^3\right) = \frac{1}{n^2} + \sum_{i=1}^{n} \varepsilon_i^2\left(\frac{3}{n} + \varepsilon_i\right).\end{aligned}$$

For a fair die this is $\frac{1}{n^2}$.

Ask yourself The cubes ε_i^3 can be negative. Could they outweigh the squares?

No, because the biases cannot be very negative: $p_i \geq 0$ gives $\varepsilon_i \geq -\frac{1}{n}$, so

$$\frac{3}{n} + \varepsilon_i \geq \frac{2}{n} > 0.$$

Each term $\varepsilon_i^2\left(\frac{3}{n} + \varepsilon_i\right)$ is therefore ≥ 0, and it is strictly positive whenever $\varepsilon_i \neq 0$, which happens for at least one i on a biased die. So

$$\sum p_i^3 > \frac{1}{n^2},$$

and a biased die is again *more* likely than a fair one to show the same score on three successive rolls.

Checks

- Take $n = 4$ with probabilities $\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$, so $\varepsilon = \frac{1}{4}, -\frac{1}{12}, -\frac{1}{12}, -\frac{1}{12}$. Directly, $\sum p_i^2 = \frac{1}{4} + \frac{3}{36} = \frac{1}{3}$, and $\frac{1}{4} + \sum \varepsilon_i^2 = \frac{1}{4} + \frac{1}{16} + \frac{3}{144} = \frac{1}{3}$, which beats the fair $\frac{1}{4}$. For three rolls, $\sum p_i^3 = \frac{1}{8} + \frac{3}{216} = \frac{5}{36} \approx 0.139$, well above the fair $\frac{1}{16} = 0.0625$.
- The most extreme die always shows 1: then both probabilities are 1, the largest possible.
- In (ii), with $n = 2$ the inequality reads $x_1 x_2 \le \frac{1}{4}(x_1 + x_2)^2$, which is AM–GM. The inequalities in (i), (ii) and (iii) were also confirmed by computer on many thousands of random cases.

Common slips

- Forgetting that $\sum \varepsilon_i = 0$, and so being left with linear terms whose sign is unknown.
- In (iii), declaring $\sum \varepsilon_i^3$ to be "small" or of either sign without using $\varepsilon_i \ge -\frac{1}{n}$; that bound is what makes the argument work.
- In (ii), applying part (i) to the x_i themselves rather than to $x_i / \sum x_j$, which need not add up to 1.

Beyond STEP

The quantity $\sum p_i^2$ is the *collision probability* of a distribution, and part (i) says it is smallest for the uniform distribution; part (iii) is the same statement for $\sum p_i^3$. Both are cases of the convexity of $t \mapsto t^k$ (Jensen's inequality), or of the power-mean inequality, and the logarithm of $\sum p_i^k$ gives the Rényi entropies used in information theory and cryptography. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet these ideas in a probabilistic setting.

WORKED SOLUTION

STEP III 2020, Question 1

Pure • Reduction Formulae

STEP III 2020, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Part (i) is a reduction formula, so integration by parts is the natural first move: differentiate the power of $\cos x$ and integrate $\cos bx$. That leaves a product $\sin x \sin bx$, and the compound-angle formula for $\cos(b-1)x$ turns it back into integrals of the original type. Part (ii) then needs nothing new: the recurrence lowers both a and b by one, which keeps the gap $b-a = 2m+1$ fixed, so it steps neatly from n to $n+1$.

Solution

Part (i)

Let $a, b \geq 1$. Integrate by parts with $u = \cos^a x$ and $\mathrm{d}v = \cos bx\,\mathrm{d}x$, so that $\mathrm{d}u = -a\cos^{a-1}x \sin x\,\mathrm{d}x$ and $v = \frac{1}{b}\sin bx$ (fine, since $b \neq 0$):

$$\mathrm{I}(a,b) = \left[\frac{\cos^a x \sin bx}{b}\right]_0^{\pi/2} + \frac{a}{b}\int_0^{\pi/2} \cos^{a-1} x \sin x \sin bx\,\mathrm{d}x.$$

The boundary term vanishes at both ends: at $x = \frac{\pi}{2}$ because $\cos^a \frac{\pi}{2} = 0$ (this is where $a \geq 1$ is used), and at $x = 0$ because $\sin 0 = 0$.

Ask yourself How can we turn $\sin x \sin bx$ back into cosines of multiples of x?

We want $\mathrm{I}(a-1, b-1)$ to appear, so look at $\cos(b-1)x$:

$$\cos(b-1)x = \cos bx \cos x + \sin bx \sin x \quad \Longrightarrow \quad \sin x \sin bx = \cos(b-1)x - \cos x \cos bx.$$

Substituting,

$$\begin{aligned}\mathrm{I}(a,b) &= \frac{a}{b}\int_0^{\pi/2} \cos^{a-1} x \cos(b-1)x\,\mathrm{d}x - \frac{a}{b}\int_0^{\pi/2} \cos^a x \cos bx\,\mathrm{d}x \\ &= \frac{a}{b}\mathrm{I}(a-1,b-1) - \frac{a}{b}\mathrm{I}(a,b).\end{aligned}$$

Multiplying by b and collecting the $\mathrm{I}(a,b)$ terms gives $(a+b)\,\mathrm{I}(a,b) = a\,\mathrm{I}(a-1,b-1)$, and since $a+b>0$,

$$\mathrm{I}(a,b) = \frac{a}{a+b}\,\mathrm{I}(a-1,b-1),$$

as required.

Part (ii)

The integral on the left is $\mathrm{I}(n,\, n+2m+1)$. Write

$$R(n,m) = (-1)^m\,\frac{2^n\,n!\,(2m)!\,(n+m)!}{m!\,(2n+2m+1)!}$$

for the right-hand side. Fix any non-negative integer m; we prove $\mathrm{I}(n, n+2m+1) = R(n,m)$ for all $n \geq 0$ by induction on n.

Base case, $n=0$.

$$\mathrm{I}(0,2m+1) = \int_0^{\pi/2} \cos(2m+1)x\,\mathrm{d}x = \frac{\sin\left((2m+1)\frac{\pi}{2}\right)}{2m+1} = \frac{(-1)^m}{2m+1},$$

because $\sin\left(m\pi+\frac{\pi}{2}\right) = \cos m\pi = (-1)^m$. On the other side,

$$R(0,m) = (-1)^m\,\frac{1\cdot 1\cdot(2m)!\,m!}{m!\,(2m+1)!} = \frac{(-1)^m}{2m+1}.$$

So the result holds for $n=0$.

Inductive step. Suppose $\mathrm{I}(k, k+2m+1) = R(k,m)$ for some $k \geq 0$. Apply part (i) with $a = k+1$ and $b = k+2m+2$, both positive integers, so that $a-1=k$, $b-1 = k+2m+1$ and $a+b = 2k+2m+3$:

$$\mathrm{I}(k+1,\, k+2m+2) = \frac{k+1}{2k+2m+3}\,\mathrm{I}(k,\, k+2m+1) = \frac{k+1}{2k+2m+3}\,R(k,m).$$

Note that $k+2m+2 = (k+1)+2m+1$, so the left-hand side is exactly the integral for $n = k+1$ with the same m.

> **?** **Ask yourself** What does the formula say $R(k+1,m)/R(k,m)$ should be?

Going from $n=k$ to $n=k+1$ in R, the factor $(-1)^m(2m)!/m!$ is unchanged, and

$$\begin{aligned}\frac{R(k+1,m)}{R(k,m)} &= \frac{2^{k+1}}{2^k}\cdot\frac{(k+1)!}{k!}\cdot\frac{(k+m+1)!}{(k+m)!}\cdot\frac{(2k+2m+1)!}{(2k+2m+3)!}\\ &= \frac{2(k+1)(k+m+1)}{(2k+2m+3)(2k+2m+2)} = \frac{k+1}{2k+2m+3},\end{aligned}$$

using $2k+2m+2 = 2(k+m+1)$. Hence

$$\mathrm{I}(k+1,\, k+2m+2) = \frac{k+1}{2k+2m+3}\,R(k,m) = R(k+1,m),$$

which is the statement for $n = k + 1$.

Since the result holds for $n = 0$, and its truth for $n = k$ implies its truth for $n = k + 1$, it holds for all non-negative integers n, by induction. As m was arbitrary, this holds for all non-negative integers m too:

$$\int_0^{\pi/2} \cos^n x \, \cos(n + 2m + 1)x \, \mathrm{d}x = (-1)^m \frac{2^n \, n! \, (2m)! \, (n + m)!}{m! \, (2n + 2m + 1)!}.$$

Checks

- Take $n = 1$, $m = 0$: the formula gives $\frac{2 \cdot 1 \cdot 1 \cdot 1}{1 \cdot 3!} = \frac{1}{3}$. Directly, $\cos x \cos 2x = \frac{1}{2}(\cos 3x + \cos x)$ integrates over $[0, \frac{\pi}{2}]$ to $\frac{1}{2}\left(-\frac{1}{3} + 1\right) = \frac{1}{3}$.
- Take $n = 1$, $m = 1$: the formula gives $-\frac{2 \cdot 2 \cdot 2}{5!} = -\frac{1}{15}$, and a numerical evaluation of $\int_0^{\pi/2} \cos x \cos 5x \, \mathrm{d}x$ gives -0.066667.
- The recurrence in (i) and the closed form in (ii) were also confirmed by computer for all $n \leq 6$, $m \leq 5$.

Common slips

- Forgetting to say why the boundary term vanishes; the top limit needs $a \geq 1$.
- Inducting on n but letting m change: part (i) lowers a and b together, so m stays fixed.
- Sign errors in $\sin\left((2m + 1)\frac{\pi}{2}\right)$; it alternates $1, -1, 1, \ldots$, giving the $(-1)^m$.

Beyond STEP

Reduction formulae like this are the integral analogue of recurrence relations, and the closed form is built from ratios of factorials in the same way as Wallis's product and the Beta function $\int_0^{\pi/2} \cos^p x \sin^q x \, \mathrm{d}x$, which you meet early in a university analysis or methods course. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2020, Question 2

Pure • Hyperbolic Curve Sketching

STEP III 2020, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Differentiate implicitly, and keep using $\cosh^2 = 1 + \sinh^2$ to turn everything back into sines: the second-derivative condition then factorises, with one factor being the curve's own equation. For the line $x + y = a$, substitute $y = a - x$ and write everything in exponentials; a real point of intersection needs a non-negative discriminant, and that gives the bound on $\cosh a$.

Solution

Part (i)

Since sinh is a bijection of $\mathbb{R}$, for each x there is exactly one y with $\sinh y = 2k - \sinh x$. So C is the graph of a function $y = \operatorname{arsinh}(2k - \sinh x)$, defined for all real x, and we may differentiate implicitly:

$$\cosh x + \cosh y \frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\cosh x}{\cosh y}.$$

Since $\cosh x \geq 1$, the gradient is never zero (it is always negative), so C has no stationary points.

Differentiating again, writing $y' = \mathrm{d}y/\mathrm{d}x$,

$$\sinh x + \sinh y\,(y')^2 + \cosh y\, y'' = 0.$$

Substitute $y' = -\cosh x / \cosh y$ and multiply through by $\cosh^2 y$:

$$\begin{aligned}
\cosh^3 y\, y'' &= -\left(\sinh x \cosh^2 y + \sinh y \cosh^2 x\right) \\
&= -\left(\sinh x\,(1 + \sinh^2 y) + \sinh y\,(1 + \sinh^2 x)\right) \\
&= -(\sinh x + \sinh y) - \sinh x \sinh y\,(\sinh y + \sinh x) \\
&= -(\sinh x + \sinh y)(1 + \sinh x \sinh y).
\end{aligned}$$

On C the first factor is $2k$, so

$$y'' = -\frac{2k\,(1+\sinh x \sinh y)}{\cosh^3 y}.$$

Since $k > 0$ and $\cosh y > 0$, this gives $y'' = 0$ if and only if $1 + \sinh x \sinh y = 0$, as required.

Ask yourself Is $y'' = 0$ enough to give a point of inflection?

No: we also need y'' to change sign. Put $s = \sinh x$ and $t = \sinh y$. At a point with $y'' = 0$ we have $s + t = 2k$ and $st = -1$, so s and t are the roots of $u^2 - 2ku - 1 = 0$:

$$u = k \pm \sqrt{k^2+1}.$$

One root is positive and one negative, and s can take either value, with t the other. Along the curve, $1 + st = 1 + s(2k - s) = -(s^2 - 2ks - 1)$ is a quadratic in s with two simple roots, so it changes sign at each one. Since $s = \sinh x$ is increasing in x, y'' does change sign at both points. The points of inflection are therefore

$$\left(\operatorname{arsinh}\left(k+\sqrt{k^2+1}\right),\ \operatorname{arsinh}\left(k-\sqrt{k^2+1}\right)\right) \quad \text{and}$$
$$\left(\operatorname{arsinh}\left(k-\sqrt{k^2+1}\right),\ \operatorname{arsinh}\left(k+\sqrt{k^2+1}\right)\right).$$

They are reflections of each other in $y = x$, which we expect, since the equation of C is symmetric in x and y.

Part (ii)

On the line, $y = a - x$. Writing $2\sinh u = \mathrm{e}^u - \mathrm{e}^{-u}$, the equation of C doubled becomes

$$\mathrm{e}^x - \mathrm{e}^{-x} + \mathrm{e}^{a-x} - \mathrm{e}^{x-a} = 4k.$$

Multiplying by e^x gives $\mathrm{e}^{2x} - 1 + \mathrm{e}^a - \mathrm{e}^{2x-a} = 4k\mathrm{e}^x$, that is,

$$\mathrm{e}^{2x}(1 - \mathrm{e}^{-a}) - 4k\mathrm{e}^x + (\mathrm{e}^a - 1) = 0,$$

which is the required equation.

Ask yourself What does the existence of such a point tell us about this equation?

It is a quadratic in $X = \mathrm{e}^x$ (provided its leading coefficient is non-zero) and it must have a real root.

First, $a \neq 0$: if $a = 0$ the equation collapses to $-4k\mathrm{e}^x = 0$, which is impossible since $k > 0$. Hence $\cosh a > 1$.

With $a \neq 0$ the leading coefficient $1 - \mathrm{e}^{-a}$ is non-zero, so the quadratic has a real root only if its discriminant is non-negative:

$$16k^2 - 4(1 - \mathrm{e}^{-a})(\mathrm{e}^a - 1) \geq 0.$$

Now $(1-\mathrm{e}^{-a})(\mathrm{e}^{a}-1) = \mathrm{e}^{a}-2+\mathrm{e}^{-a} = 2\cosh a - 2$, so the condition is $16k^2 \geq 8(\cosh a - 1)$, i.e. $\cosh a \leq 2k^2+1$. Together,

$$1 < \cosh a \leq 2k^2+1.$$

So every point of C has $x+y$ between $-2\operatorname{arsinh} k$ and $2\operatorname{arsinh} k$ (because $\cosh(2\operatorname{arsinh} k) = 1+2\sinh^2(\operatorname{arsinh} k) = 1+2k^2$), and $x+y \neq 0$.

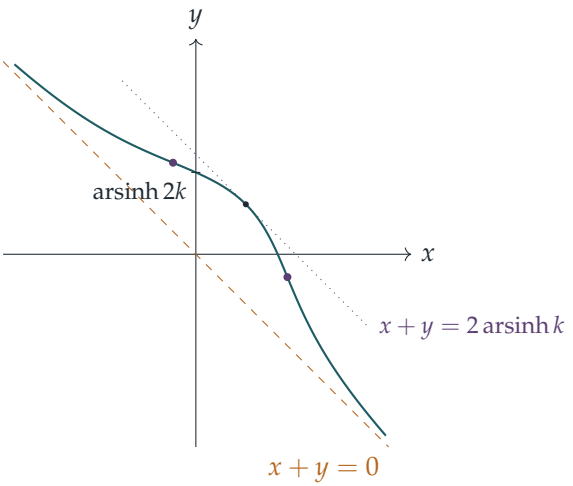
In fact $x+y > 0$ on C. If $a < 0$ then $1-\mathrm{e}^{-a} < 0$ and $\mathrm{e}^{a}-1 < 0$, so all three coefficients of the quadratic are negative and it has no positive root X. (Equivalently, $\sinh x + \sinh y = 2\sinh\frac{x+y}{2}\cosh\frac{x-y}{2}$, which has the sign of $x+y$.) The largest value $x+y = 2\operatorname{arsinh} k$ occurs at $x = y = \operatorname{arsinh} k$, where the discriminant vanishes and the line just touches C.

Part (iii)

Collecting what we know:

- C is symmetric in the line $y = x$ and crosses it only at $(\operatorname{arsinh} k, \operatorname{arsinh} k)$, where the gradient is -1.
- y is defined for all x and the gradient is always negative. The curve meets the axes at $(\operatorname{arsinh} 2k, 0)$ and $(0, \operatorname{arsinh} 2k)$.
- $0 < x+y \leq 2\operatorname{arsinh} k$, so C lies in the strip above the line $x+y=0$ and on or below the line $x+y = 2\operatorname{arsinh} k$, which touches C at its point on $y = x$.
- As $x \to +\infty$, $\sinh y = 2k - \sinh x \to -\infty$, and $\mathrm{e}^{-y} \approx \mathrm{e}^{x} - 4k$, so $x+y \approx -\ln(1-4k\mathrm{e}^{-x}) \approx 4k\mathrm{e}^{-x} \to 0^{+}$. By symmetry the same happens as $x \to -\infty$. So $x+y=0$ is an asymptote at both ends, approached from above.
- By the formula for y'', the curve is concave (below its tangents) between the two points of inflection, where $1+\sinh x \sinh y > 0$ (for example at the point on $y = x$), and convex outside them.

The sketch shows the case $k = 1$, with the points of inflection marked.



Checks

- With $k = 1$ the points of inflection are at $\left(\operatorname{arsinh}(1+\sqrt{2}), \operatorname{arsinh}(1-\sqrt{2})\right) \approx (1.615, -0.403)$ and its mirror image; checking second differences of $y = \operatorname{arsinh}(2 - \sinh x)$ on either side confirms y'' changes sign there, and a scan over $-8 \le x \le 8$ finds no other zeros of y''.
- The formula $y'' = -2k(1 + \sinh x \sinh y)/\cosh^3 y$ agrees with numerical second derivatives at many random points and for several values of k.
- For $k = 1$ and $x = 6$, $x + y \approx 0.00996$, close to the predicted $4\mathrm{e}^{-6} \approx 0.00992$: the asymptote is approached from above, as claimed.
- At $x = y = \operatorname{arsinh} k$, $\cosh(x+y) = 1 + 2\sinh^2 x = 1 + 2k^2$: the upper bound in (ii) is attained.

Common slips

- Stopping at $y'' = 0$ without checking that y'' changes sign. Here it does, but it needs saying.
- Forgetting to rule out $a = 0$, which is what makes the left-hand inequality strict.
- Only writing down one point of inflection. Either root of $u^2 - 2ku - 1 = 0$ can be $\sinh x$, giving two points mirrored in $y = x$.

Beyond STEP

The rotated coordinates $u = x + y$, $v = x - y$ make this curve transparent: the identity $\sinh x + \sinh y = 2\sinh\frac{u}{2}\cosh\frac{v}{2}$ turns C into $\sinh\frac{u}{2} = k\operatorname{sech}\frac{v}{2}$, a single hump over the asymptote $u = 0$. Choosing coordinates adapted to a symmetry is a theme throughout

first-year calculus and linear algebra (diagonalising a quadratic form is the same idea). Implicit curves, and when an equation $F(x, y) = 0$ really defines y as a smooth function of x, are the subject of the implicit function theorem; Bartle and Sherbert, *Introduction to Real Analysis*, gives a careful treatment of the one-variable analysis behind it.

WORKED SOLUTION

STEP III 2020, Question 3

Pure • Complex Numbers in Geometry

STEP III 2020, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

In the complex plane a rotation about a through angle θ is just $z \mapsto a + (z-a)e^{i\theta}$, so G_{AB} is an explicit linear combination of a and b. Once part (i) gives that formula, the geometry turns into algebra: a parallelogram is a linear condition on the vertices, and an equilateral triangle is one side being another side turned through a fixed angle. The whole question rests on a few facts about powers of ω.

Solution

Throughout, $\omega = e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$, so $\omega^* = e^{-i\pi/6}$, and we write

$$\rho = e^{-i\pi/3} = \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i.$$

Three facts we shall use:

$$\omega^2 = e^{i\pi/3}, \qquad \omega^3 = i, \qquad \omega - \omega^* = 2i\sin\tfrac{\pi}{6} = i.$$

Part (i)

A clockwise rotation through $\frac{1}{3}\pi$ multiplies by ρ. Rotating about A means we rotate the displacement $b-a$ and add it back on to a, so

$$k = a + (b-a)\rho.$$

The centroid of ABK is therefore

$$g_{ab} = \tfrac{1}{3}(a+b+k) = \tfrac{1}{3}\left[(2-\rho)\,a + (1+\rho)\,b\right].$$

Now we tidy the two coefficients:

$$2-\rho = \tfrac{3}{2} + \tfrac{\sqrt{3}}{2}i = \sqrt{3}\left(\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}i\right) = \sqrt{3}\,\omega,$$
$$1+\rho = \tfrac{3}{2} - \tfrac{\sqrt{3}}{2}i = \sqrt{3}\left(\tfrac{\sqrt{3}}{2} - \tfrac{1}{2}i\right) = \sqrt{3}\,\omega^*.$$

Hence

$$g_{ab} = \tfrac{1}{3}\left(\sqrt{3}\,\omega a + \sqrt{3}\,\omega^* b\right) = \frac{1}{\sqrt{3}}\left(\omega a + \omega^* b\right),$$

as required.

Part (ii)

A quadrilateral $PQRS$ (vertices in that order) is a parallelogram exactly when $\overrightarrow{PQ} = \overrightarrow{SR}$, that is, $q - p = r - s$, or equivalently

$$p - q + r - s = 0.$$

(This is the same as saying the diagonals PR and QS have the same midpoint.) So Q_1 is a parallelogram if and only if $a - b + c - d = 0$, and Q_2 is a parallelogram if and only if

$$g_{ab} - g_{bc} + g_{cd} - g_{da} = 0.$$

Using part (i) for each of the four points,

$$\begin{aligned}\sqrt{3}\left(g_{ab} - g_{bc} + g_{cd} - g_{da}\right) &= (\omega a + \omega^* b) - (\omega b + \omega^* c) + (\omega c + \omega^* d) - (\omega d + \omega^* a)\\ &= (\omega - \omega^*)a - (\omega - \omega^*)b + (\omega - \omega^*)c - (\omega - \omega^*)d\\ &= \mathrm{i}\,(a - b + c - d).\end{aligned}$$

Thus

$$g_{ab} - g_{bc} + g_{cd} - g_{da} = \frac{\mathrm{i}}{\sqrt{3}}\,(a - b + c - d),$$

and since $\mathrm{i}/\sqrt{3} \neq 0$, one side vanishes if and only if the other does. Therefore Q_1 is a parallelogram if and only if Q_2 is.

Ask yourself Could Q_2 satisfy the parallelogram condition but collapse, with two of its vertices equal?

Yes, in one special case. If ABC is an equilateral triangle labelled clockwise, then rotating B clockwise about A lands on C, so G_{AB} and G_{BC} are both the centroid of ABC. Completing the rhombus $ABCD$, Q_2 flattens to a segment. The condition $p - q + r - s = 0$ still holds there, so the equivalence is really a statement about this vector condition, which is what "parallelogram" means in a question like this.

Part (iii)

We show that one side of T_2, as a vector, is another side turned through a fixed angle. From part (i),

$$\begin{aligned}\sqrt{3}\left(g_{bc} - g_{ab}\right) &= (\omega b + \omega^* c) - (\omega a + \omega^* b) = -\omega a + \mathrm{i}\,b + \omega^* c,\\ \sqrt{3}\left(g_{ca} - g_{bc}\right) &= (\omega c + \omega^* a) - (\omega b + \omega^* c) = \omega^* a - \omega b + \mathrm{i}\,c,\end{aligned}$$

using $\omega - \omega^* = \mathrm{i}$ in each line.

Ask yourself What number λ would turn the second bracket into the first?

Matching the coefficients of a forces $\lambda\omega^* = -\omega$, so $\lambda = -\omega/\omega^* = -\omega^2 = -\mathrm{e}^{\mathrm{i}\pi/3} = \mathrm{e}^{-2\pi\mathrm{i}/3}$. This same λ works for b and c too:

$$\lambda \cdot (-\omega) = \omega^3 = \mathrm{i},$$
$$\lambda \cdot \mathrm{i} = -\omega^2 \cdot \omega^3 = -\omega^5 = -\mathrm{e}^{5\pi\mathrm{i}/6} = \mathrm{e}^{-\mathrm{i}\pi/6} = \omega^*.$$

Hence, for all a, b, c,

$$g_{bc} - g_{ab} = \mathrm{e}^{-2\pi\mathrm{i}/3}\,(g_{ca} - g_{bc}).$$

Taking moduli, $G_{AB}G_{BC} = G_{BC}G_{CA}$, since $\left|\mathrm{e}^{-2\pi\mathrm{i}/3}\right| = 1$. Also the direction of travel along T_2 turns through $\frac{2}{3}\pi$ at G_{BC}, so the interior angle of T_2 at G_{BC} is $\pi - \frac{2}{3}\pi = \frac{1}{3}\pi$. A triangle with two equal sides enclosing an angle of $\frac{1}{3}\pi$ is isosceles with base angles $\frac{1}{2}(\pi - \frac{1}{3}\pi) = \frac{1}{3}\pi$, so all its angles are $\frac{1}{3}\pi$ and T_2 is equilateral.

The one exception is when $g_{ca} = g_{bc}$, and then the relation gives $g_{ab} = g_{bc}$ as well, so T_2 shrinks to a single point. Multiplying $\omega^* a - \omega b + \mathrm{i}c = 0$ by ω gives $a - \mathrm{e}^{\mathrm{i}\pi/3}b + \mathrm{e}^{2\pi\mathrm{i}/3}c = 0$, which happens exactly when ABC is itself an equilateral triangle labelled clockwise (as in the remark in part (ii)). Every other triangle T_1, including degenerate ones, gives a genuine equilateral T_2.

Checks

- Take $a = 0$, $b = 1$. Then $k = \rho$, and the centroid of $0, 1, \rho$ is $\frac{1}{3}(1 + \rho) = \frac{1}{2} - \frac{\sqrt{3}}{6}\mathrm{i}$. The formula gives $\omega^*/\sqrt{3} = \frac{1}{2} - \frac{1}{2\sqrt{3}}\mathrm{i}$, the same number.
- With $a = 0$, $b = 1$, $c = \mathrm{i}$ (a right-angled triangle), the three points of T_2 have pairwise distances all equal to about 1.1154. A computer check on 10^4 random triangles and quadrilaterals confirmed the identities in parts (ii) and (iii) to rounding error.

Common slips

- Rotating b about the origin, writing $k = b\rho$, rather than about A.
- Using $\mathrm{e}^{+\mathrm{i}\pi/3}$ for a clockwise rotation. Clockwise is the negative sense.
- In part (ii), proving only one direction. The factor $\mathrm{i}/\sqrt{3}$ being non-zero is what gives "if and only if".

Beyond STEP

Part (iii) is Napoleon's theorem. Erect equilateral triangles on the sides of any triangle (here outwards, when ABC is labelled anticlockwise), and their centroids form an equilateral triangle. The complex-number proof works because rotations and centroids are both linear maps of the plane. That is the starting point for the study of transformation

groups in geometry. Needham's *Visual Complex Analysis* treats this geometric view of complex multiplication beautifully.

WORKED SOLUTION

STEP III 2020, Question 4

Pure • Reflection and Rotation Matrices

STEP III 2020, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The opening result is the whole question in disguise. Subtracting $(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$ takes a point straight down to the plane, so subtracting *twice* that amount carries it the same distance out the other side, which is a reflection. Once a reflection is written as $\mathbf{x}\mapsto\mathbf{x}-2(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$, its matrix, a half-turn about $\mathbf{n}$, and their product all follow with very little calculation.

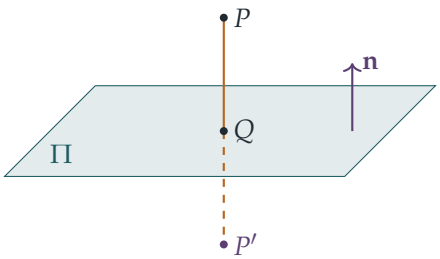
Solution

The opening result

Let $\mathbf{q}=\mathbf{x}-(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$. Since $\mathbf{n}\cdot\mathbf{n}=1$,

$$\mathbf{q}\cdot\mathbf{n}=\mathbf{x}\cdot\mathbf{n}-(\mathbf{x}\cdot\mathbf{n})(\mathbf{n}\cdot\mathbf{n})=0,$$

so Q lies on Π. Also $\overrightarrow{PQ}=\mathbf{q}-\mathbf{x}=-(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$, which is a multiple of the normal $\mathbf{n}$. It is non-zero because P is not on Π (so $\mathbf{x}\cdot\mathbf{n}\neq 0$). Hence PQ is perpendicular to Π, as required.



The picture also shows where we are heading: the mirror image P' of P in Π is as far beyond Q as P is above it, so its position vector is

$$\mathbf{x}-2(\mathbf{x}\cdot\mathbf{n})\mathbf{n}.$$

(If P is on Π this formula gives P itself, which is also correct.)

Part (i)

Take $\mathbf{n} = (a, b, c)^{\mathsf{T}}$, a unit normal because $a^2 + b^2 + c^2 = 1$. For $\mathbf{i}$ we have $\mathbf{i} \cdot \mathbf{n} = a$, so

$$T(\mathbf{i}) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 2a \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 - 2a^2 \\ -2ab \\ -2ac \end{pmatrix} = \begin{pmatrix} b^2 + c^2 - a^2 \\ -2ab \\ -2ac \end{pmatrix},$$

using $1 = a^2 + b^2 + c^2$ in the top entry. This is the required image. In the same way, with $\mathbf{j} \cdot \mathbf{n} = b$ and $\mathbf{k} \cdot \mathbf{n} = c$,

$$T(\mathbf{j}) = \begin{pmatrix} -2ab \\ a^2 + c^2 - b^2 \\ -2bc \end{pmatrix}, \qquad T(\mathbf{k}) = \begin{pmatrix} -2ac \\ -2bc \\ a^2 + b^2 - c^2 \end{pmatrix}.$$

The columns of $\mathbf{M}$ are the images of $\mathbf{i}, \mathbf{j}, \mathbf{k}$, so

$$\mathbf{M} = \begin{pmatrix} b^2 + c^2 - a^2 & -2ab & -2ac \\ -2ab & a^2 + c^2 - b^2 & -2bc \\ -2ac & -2bc & a^2 + b^2 - c^2 \end{pmatrix} = \mathbf{I} - 2\mathbf{n}\mathbf{n}^{\mathsf{T}}.$$

Part (ii)

We match the given matrix with $\mathbf{M}$, writing each diagonal entry as $1 - 2a^2$, $1 - 2b^2$, $1 - 2c^2$:

$$1 - 2a^2 = 0.64, \quad 1 - 2b^2 = 0.36, \quad 1 - 2c^2 = 0 \implies a^2 = 0.18, \quad b^2 = 0.32, \quad c^2 = 0.5.$$

So $|a| = \frac{3}{5\sqrt{2}}$, $|b| = \frac{4}{5\sqrt{2}}$ and $|c| = \frac{5}{5\sqrt{2}}$.

? **Ask yourself** The diagonal only gives a^2, b^2, c^2. How do we settle the signs?

The off-diagonal entries give the products:

$$-2ab = 0.48, \quad -2ac = 0.6, \quad -2bc = -0.8 \implies ab = -0.24, \quad ac = -0.3, \quad bc = 0.4.$$

So b and c have the same sign, and a has the opposite sign. Replacing (a, b, c) by $(-a, -b, -c)$ changes nothing (it is the same plane), so we may take $c > 0$, which gives

$$(a, b, c) = \tfrac{1}{5\sqrt{2}}(-3,\ 4,\ 5).$$

These values give $ab = -\frac{12}{50}$, $ac = -\frac{15}{50}$ and $bc = \frac{20}{50}$, so all six independent entries match. Multiplying through by $-5\sqrt{2}$, the plane is

$$3x - 4y - 5z = 0.$$

Part (iii)

Split any vector as $\mathbf{x} = (\mathbf{x} \cdot \mathbf{n})\mathbf{n} + \big(\mathbf{x} - (\mathbf{x} \cdot \mathbf{n})\mathbf{n}\big)$, a part along the axis plus a part perpendicular to it (the opening result). A rotation through π about the axis leaves the

first part alone and reverses the second, so

$$\mathbf{x} \mapsto (\mathbf{x}\cdot\mathbf{n})\mathbf{n} - \big(\mathbf{x} - (\mathbf{x}\cdot\mathbf{n})\mathbf{n}\big) = 2(\mathbf{x}\cdot\mathbf{n})\mathbf{n} - \mathbf{x}.$$

This is exactly $-T(\mathbf{x})$. Therefore $\mathbf{N} = 2\mathbf{n}\mathbf{n}^{\mathsf{T}} - \mathbf{I} = -\mathbf{M}$, that is,

$$\mathbf{N} = \begin{pmatrix} a^2-b^2-c^2 & 2ab & 2ac \\ 2ab & b^2-a^2-c^2 & 2bc \\ 2ac & 2bc & c^2-a^2-b^2 \end{pmatrix}.$$

Part (iv)

Ask yourself Do we need to multiply two 3×3 matrices of a, b, c here?

No. From (iii), $\mathbf{NM} = -\mathbf{M}^2$. Reflecting twice in the same plane brings every point back to where it started, so $\mathbf{M}^2 = \mathbf{I}$. (Algebraically, $(\mathbf{I} - 2\mathbf{n}\mathbf{n}^{\mathsf{T}})^2 = \mathbf{I} - 4\mathbf{n}\mathbf{n}^{\mathsf{T}} + 4\mathbf{n}(\mathbf{n}^{\mathsf{T}}\mathbf{n})\mathbf{n}^{\mathsf{T}} = \mathbf{I}$, because $\mathbf{n}^{\mathsf{T}}\mathbf{n} = 1$.) Hence

$$\mathbf{NM} = -\mathbf{I},$$

which sends every $\mathbf{x}$ to $-\mathbf{x}$. This is reflection in the origin, also called central inversion or an enlargement with scale factor -1 centred at the origin. Notice that it doesn't depend on a, b or c at all.

Checks

- Each column of $\mathbf{M}$ should be a unit vector. For the first, $(b^2+c^2-a^2)^2 + 4a^2b^2 + 4a^2c^2 = (b^2+c^2+a^2)^2 = 1$.
- In (ii), the normal $(3,-4,-5)$ should be sent to its negative. The given matrix maps it to $(1.92 - 1.92 - 3,\ 1.44 - 1.44 + 4,\ 1.8 + 3.2) = (-3,4,5)$. Good.
- Taking $\mathbf{n} = \mathbf{k}$ makes $\mathbf{M} = \operatorname{diag}(1,1,-1)$, reflection in $z = 0$, and $\mathbf{N} = \operatorname{diag}(-1,-1,1)$, the half-turn about the z-axis. Their product is clearly $-\mathbf{I}$.
- A computer check with 1000 random unit vectors $\mathbf{n}$ confirmed the formula for $\mathbf{M}$, that $\mathbf{N}$ fixes $\mathbf{n}$ and reverses perpendicular vectors, and that $\mathbf{NM} = -\mathbf{I}$. The matrix in (ii) was reproduced to within 10^{-15}.

Common slips

- Using $\mathbf{x} - (\mathbf{x}\cdot\mathbf{n})\mathbf{n}$ as the reflection. That is the foot of the perpendicular, so the factor 2 is missing.
- In (ii), choosing the signs of a, b, c independently from the square roots. Only the off-diagonal entries fix the relative signs.
- Saying $\mathbf{NM}$ is "the identity". A reflection has determinant -1 and a rotation has determinant $+1$, so the product has determinant -1. In three dimensions $\det(-\mathbf{I}) = -1$, which is consistent.

Beyond STEP

The matrix $\mathbf{I} - 2\mathbf{n}\mathbf{n}^{\mathsf{T}}$ is a *Householder reflection*. Numerical linear algebra uses it constantly to reduce matrices to triangular form, and every orthogonal matrix in n dimensions is a product of at most n such reflections. Part (iv) is a small instance of that structure: in $\mathbb{R}^3$, a half-turn composed with the reflection in the perpendicular plane gives $-\mathbf{I}$. For orthogonal matrices, projections and reflections, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 2020, Question 5

Pure • Partial Fractions

STEP III 2020, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The opening identity is the finite geometric series in disguise, and it is exactly what turns $\left(1-(x/k)^n\right)/(x-k)$ into an explicit polynomial. That gives a complete partial-fraction decomposition of $1/\left(x^n(x-k)\right)$ without any heavy comparing of coefficients. Differentiating then produces the decomposition with a *repeated* factor $(x-k)^2$, and the final integral is just a special case in which every term can be integrated directly.

Solution

The opening identity

Expand the right-hand side and split it into two sums:

$$\begin{aligned}(x-y)\sum_{r=1}^{n} x^{n-r}y^{r-1} &= \sum_{r=1}^{n} x^{n-r+1}y^{r-1} - \sum_{r=1}^{n} x^{n-r}y^{r} \\ &= \left(x^n + x^{n-1}y + \cdots + xy^{n-1}\right) - \left(x^{n-1}y + \cdots + xy^{n-1} + y^n\right) \\ &= x^n - y^n,\end{aligned}$$

since every term except x^n (from $r=1$ in the first sum) and y^n (from $r=n$ in the second) appears once in each bracket and cancels.

Part (i)(a)

Multiplying $\mathrm{F}(x) = \dfrac{A}{x-k} + \dfrac{\mathrm{f}(x)}{x^n}$ through by $x^n(x-k)$ gives

$$1 = Ax^n + \mathrm{f}(x)(x-k).$$

This is an identity between polynomials, true for all $x \neq 0, k$ and hence for all x (two polynomials that agree at infinitely many points are equal). Putting $x=k$ gives $1 = Ak^n$,

so $A = 1/k^n$ (allowed, since $k \neq 0$). Then

$$\mathrm{f}(x)(x-k) = 1 - \frac{x^n}{k^n}, \qquad \text{so} \qquad \mathrm{f}(x) = \frac{1}{x-k}\left(1 - \left(\frac{x}{k}\right)^n\right)$$

for $x \neq k$, as required.

Ask yourself How do we turn this into something without $x-k$ in the denominator?

Use the opening identity with 1 in place of x and x/k in place of y:

$$1 - \left(\frac{x}{k}\right)^n = \left(1 - \frac{x}{k}\right)\sum_{r=1}^{n}\left(\frac{x}{k}\right)^{r-1} = -\frac{x-k}{k}\sum_{r=1}^{n}\frac{x^{r-1}}{k^{r-1}}.$$

Dividing by $x-k$,

$$\mathrm{f}(x) = -\frac{1}{k}\sum_{r=1}^{n}\frac{x^{r-1}}{k^{r-1}}, \qquad \text{so} \qquad \frac{\mathrm{f}(x)}{x^n} = -\frac{1}{k}\sum_{r=1}^{n}\frac{1}{k^{r-1}x^{n+1-r}}.$$

(This also confirms that f really is a polynomial.) Now re-index with $s = n+1-r$, which runs over $1,\ldots,n$ as r does; then $r-1 = n-s$, and the sum becomes $\sum_{s=1}^{n} 1/(k^{n-s}x^s)$. Therefore

$$\mathrm{F}(x) = \frac{1}{k^n(x-k)} - \frac{1}{k}\sum_{r=1}^{n}\frac{1}{k^{n-r}x^r}.$$

Part (i)(b)

Since $x^n\mathrm{F}(x) = \dfrac{1}{x-k}$, differentiating both sides with the product rule gives

$$nx^{n-1}\mathrm{F}(x) + x^n\mathrm{F}'(x) = -\frac{1}{(x-k)^2}.$$

Dividing by $-x^n$ (fine, as $x \neq 0$):

$$\frac{1}{x^n(x-k)^2} = -\frac{n}{x}\mathrm{F}(x) - \mathrm{F}'(x).$$

From part (a),

$$-\frac{n}{x}\mathrm{F}(x) = -\frac{n}{xk^n(x-k)} + \sum_{r=1}^{n}\frac{n}{k^{n+1-r}x^{r+1}},$$

and differentiating the result of part (a) term by term,

$$-\mathrm{F}'(x) = \frac{1}{k^n(x-k)^2} - \frac{1}{k}\sum_{r=1}^{n}\frac{r}{k^{n-r}x^{r+1}} = \frac{1}{k^n(x-k)^2} - \sum_{r=1}^{n}\frac{r}{k^{n+1-r}x^{r+1}}.$$

Adding, the two sums combine to give

$$\frac{1}{x^n(x-k)^2} = \frac{1}{k^n(x-k)^2} - \frac{n}{xk^n(x-k)} + \sum_{r=1}^{n}\frac{n-r}{k^{n+1-r}x^{r+1}},$$

which is the required result.

Part (ii)

Take $n = 3$ and $k = 1$ in part (b). The $r = 3$ term of the sum vanishes, and

$$\frac{1}{x^3(x-1)^2} = \frac{1}{(x-1)^2} - \frac{3}{x(x-1)} + \frac{2}{x^2} + \frac{1}{x^3}.$$

The middle term needs one more split: $\dfrac{1}{x(x-1)} = \dfrac{1}{x-1} - \dfrac{1}{x}$. On $[2, N]$ everything is continuous and $x - 1 > 0$, so an antiderivative there is

$$G(x) = -\frac{1}{x-1} - 3\ln\left(\frac{x-1}{x}\right) - \frac{2}{x} - \frac{1}{2x^2},$$

and

$$\int_2^N \frac{1}{x^3(x-1)^2}\,\mathrm{d}x = G(N) - G(2).$$

? **Ask yourself** Why must we keep the two logarithms together?

Separately, $-3\ln(N-1)$ and $+3\ln N$ each tend to infinity, so we can't take the limit term by term. Combined, $\ln\big((N-1)/N\big) = \ln\big(1 - \frac{1}{N}\big) \to \ln 1 = 0$ as $N \to \infty$, because ln is continuous at 1. The other three terms of $G(N)$ clearly tend to 0, so $G(N) \to 0$.

Finally,

$$G(2) = -1 - 3\ln\tfrac{1}{2} - 1 - \tfrac{1}{8} = 3\ln 2 - \tfrac{17}{8},$$

so

$$\lim_{N\to\infty} \int_2^N \frac{1}{x^3(x-1)^2}\,\mathrm{d}x = \frac{17}{8} - 3\ln 2.$$

Checks

- The answer must be positive, since the integrand is positive: indeed $\frac{17}{8} - 3\ln 2 \approx 2.1250 - 2.0794 = 0.0456$.
- It should also be small: the integrand is $\frac{1}{8}$ at $x = 2$ and falls off roughly like x^{-5}. Crudely bounding it by $\frac{1}{8}(2/x)^5$ gives at most $\int_2^\infty 4x^{-5}\,\mathrm{d}x = \frac{1}{16}$. A numerical integration confirms the value $0.045558\ldots$ to many decimal places.
- Try $n = 1$ in part (b). The sum vanishes (its only term has $n - r = 0$), and the right-hand side is $\dfrac{1}{k(x-k)^2} - \dfrac{1}{xk(x-k)} = \dfrac{x-(x-k)}{xk(x-k)^2} = \dfrac{1}{x(x-k)^2}$, as it should be.

Common slips

- Getting the sign wrong when writing $1 - x/k$ as $-(x-k)/k$.

- Forgetting to re-index the sum in part (a), and ending up with powers of k and x that do not match the target.
- Splitting $\ln\big((x-1)/x\big)$ into two logarithms and then claiming that the integral diverges, or writing "$\infty - \infty = 0$".

Beyond STEP

What you have built is the general partial-fraction decomposition for factors of the form x^n and $(x-k)^m$; differentiating with respect to x (or with respect to the root k) is a standard way to generate the repeated-factor terms. Deciding whether an integral over an infinite range converges, as in part (ii), is the subject of improper integrals in a first analysis course. Abbott's *Understanding Analysis* treats these limits carefully.

WORKED SOLUTION

STEP III 2020, Question 6

Pure • Polar Curves and Area

STEP III 2020, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

All three parts are built on the same function $\cos\theta + \sqrt{\cos 2\theta}$. The equation of C_2 is a quadratic in r, and its two roots are $\cos\theta \pm \sqrt{\cos 2\theta}$. So C_2 is C_1 (the larger root) together with a second, inner branch (the smaller root), and the two branches meet where the discriminant $\cos 2\theta$ vanishes. With that in place, the area between the branches collapses to a single integral, because $r_+^2 - r_-^2$ factorises.

Solution

Part (i)

Let $y = \cos x + \sqrt{\cos 2x}$. This is real exactly when $\cos 2x \geq 0$, which on the given interval is the whole of $[-\frac{1}{4}\pi, \frac{1}{4}\pi]$. The function is even, so the graph is symmetric about the y-axis.

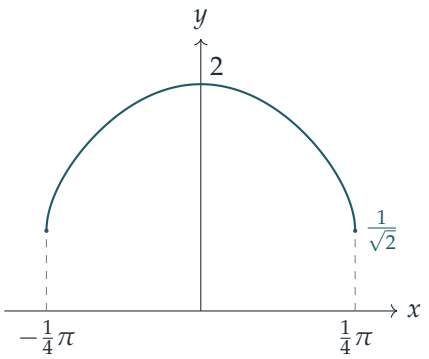
Values. At $x = 0$, $y = 2$. At $x = \pm\frac{1}{4}\pi$, $\cos 2x = 0$ and $y = \cos\frac{1}{4}\pi = \frac{1}{\sqrt{2}}$.

Gradient. For $0 < x < \frac{1}{4}\pi$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\sin x - \frac{\sin 2x}{\sqrt{\cos 2x}} < 0,$$

since both terms are negative there. So y decreases from 2 to $\frac{1}{\sqrt{2}}$ as x goes from 0 to $\frac{1}{4}\pi$, and the maximum is at $(0,2)$. As $x \to \frac{1}{4}\pi^-$, $\sqrt{\cos 2x} \to 0^+$ while $\sin 2x \to 1$, so $\mathrm{d}y/\mathrm{d}x \to -\infty$: the curve meets the end points with *vertical* tangents.

Near $x = 0$. Using $\cos x \approx 1 - \frac{1}{2}x^2$ and $\sqrt{\cos 2x} \approx \sqrt{1 - 2x^2} \approx 1 - x^2$, we get $y \approx 2 - \frac{3}{2}x^2$, a smooth maximum.



Part (ii)

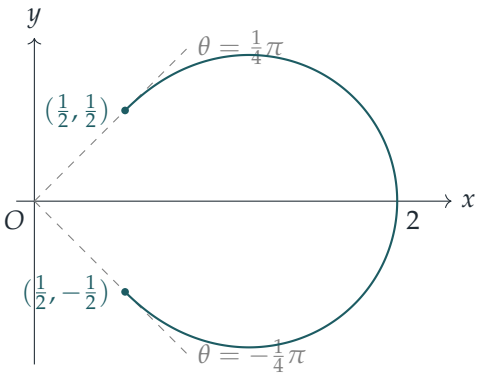
Now the same function gives the distance r from the origin in direction θ. The end points of C_1 are at $r = \frac{1}{\sqrt{2}}$, $\theta = \pm\frac{1}{4}\pi$, which are the Cartesian points $(\frac{1}{2}, \pm\frac{1}{2})$. At $\theta = 0$ the curve is at $(2, 0)$, the furthest point from O. As r is even in θ, C_1 is symmetric in the initial line.

Ask yourself What does an infinite $\mathrm{d}r/\mathrm{d}\theta$ mean for a polar curve?

From part (i), $\mathrm{d}r/\mathrm{d}\theta \to -\infty$ as $\theta \to \frac{1}{4}\pi^-$. Near that end, r changes a lot while θ hardly changes at all, so the curve is moving along the radial line $\theta = \frac{1}{4}\pi$ (the line $y = x$). Since r is decreasing, it arrives at $(\frac{1}{2}, \frac{1}{2})$ heading *towards* the origin along $y = x$. In particular, the height $y = r\sin\theta$ must reach a maximum before the end point. Indeed

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta$$

is positive near $\theta = 0$ and tends to $-\infty$ as $\theta \to \frac{1}{4}\pi$. (The maximum height is about 0.81, at $\theta \approx 34°$.) So C_1 is an arch through $(2, 0)$ that bulges above $y = \frac{1}{2}$ and comes back down to $(\frac{1}{2}, \frac{1}{2})$, touching the line $y = x$ there, with a mirror image below the axis.



Part (iii)

The two branches. Treat the equation as a quadratic in r:

$$r^2 - 2r\cos\theta + \sin^2\theta = 0 \iff r = \cos\theta \pm \sqrt{\cos^2\theta - \sin^2\theta} = \cos\theta \pm \sqrt{\cos 2\theta}.$$

Write $r_\pm = \cos\theta \pm \sqrt{\cos 2\theta}$. The outer branch r_+ is exactly C_1. The inner branch satisfies $r_- \geq 0$, because $\cos^2\theta \geq \cos^2\theta - \sin^2\theta = \cos 2\theta$, with equality only when $\sin\theta = 0$. So r_- is a genuine polar curve, and it reaches the origin only at $\theta = 0$.

At $\theta = \pm\frac{1}{4}\pi$. Here $\cos 2\theta = 0$, so the quadratic has a repeated root:

$$r = \cos\tfrac{1}{4}\pi = \frac{1}{\sqrt{2}}.$$

Both branches pass through $(\frac{1}{2}, \pm\frac{1}{2})$.

Small r. If r is small, then r^2 is negligible next to $2r\cos\theta$, and θ must be small too, since $\sin^2\theta = 2r\cos\theta - r^2$ is small. Then $2r\cos\theta \approx \sin^2\theta$, and using $\cos\theta \approx 1$, $\sin\theta \approx \theta$,

$$r \approx \frac{\sin^2\theta}{2\cos\theta} \approx \tfrac{1}{2}\theta^2,$$

as required. (Equivalently, $r_- \approx (1 - \frac{1}{2}\theta^2) - (1 - \theta^2) = \frac{1}{2}\theta^2$.)

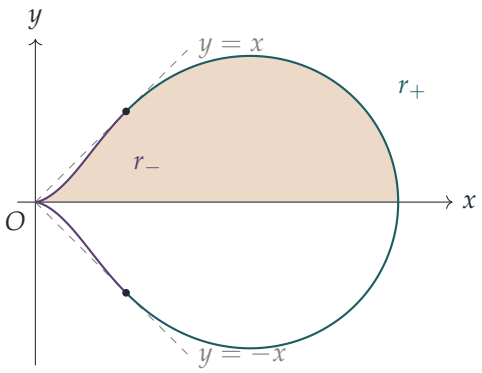
Behaviour near $r = 0$. Here $x = r\cos\theta \approx \frac{1}{2}\theta^2$ and $y = r\sin\theta \approx \frac{1}{2}\theta^3$, so $y^2 \approx 2x^3$. The inner branch therefore comes into the origin tangent to the initial line from the side $x > 0$, from above and below, forming a *cusp* at O that points left.

Behaviour near $\theta = \pm\frac{1}{4}\pi$. Differentiating the equation implicitly,

$$(2r - 2\cos\theta)\,\mathrm{d}r + (2r\sin\theta + 2\sin\theta\cos\theta)\,\mathrm{d}\theta = 0 \quad\Longrightarrow\quad \frac{\mathrm{d}\theta}{\mathrm{d}r} = \frac{\cos\theta - r}{\sin\theta\,(r + \cos\theta)}.$$

At $r = \frac{1}{\sqrt{2}}$, $\theta = \frac{1}{4}\pi$ the numerator is 0 and the denominator is not, so $\mathrm{d}\theta/\mathrm{d}r = 0$. Also $\theta \leq \frac{1}{4}\pi$ on the whole curve. So θ has a maximum there, and the curve is tangent to the radial line $y = x$. The outer branch comes in along $y = x$ towards O, the inner branch leaves along $y = x$ away from O, and the two join smoothly at $(\frac{1}{2}, \frac{1}{2})$. There is no corner. The same happens at $(\frac{1}{2}, -\frac{1}{2})$ by symmetry.

So C_2 is a closed curve, symmetric in the initial line, shaped like a teardrop whose point is the cusp at O. Its outer edge is C_1, and its inner edge runs from $(\frac{1}{2}, \frac{1}{2})$ down into the cusp at O and out again to $(\frac{1}{2}, -\frac{1}{2})$.



(The plotting uses $\sin\theta = \frac{1}{\sqrt{2}}\sin u$, which makes $\sqrt{\cos 2\theta} = \cos u$. Letting u run once round traces the whole of C_2 without any infinite slopes.)

The area. For $0 \le \theta \le \frac{1}{4}\pi$, the ray at angle θ is inside C_2 exactly for $r_- \le r \le r_+$. So the shaded area is the difference of two polar sectors:

$$\begin{aligned} A &= \int_0^{\pi/4} \tfrac{1}{2}\left(r_+^2 - r_-^2\right) \mathrm{d}\theta = \int_0^{\pi/4} \tfrac{1}{2}(r_+ - r_-)(r_+ + r_-)\,\mathrm{d}\theta \\ &= \int_0^{\pi/4} \tfrac{1}{2}\cdot 2\sqrt{\cos 2\theta}\cdot 2\cos\theta\,\mathrm{d}\theta = 2\int_0^{\pi/4} \cos\theta\sqrt{1-2\sin^2\theta}\,\mathrm{d}\theta. \end{aligned}$$

Substitute $\sqrt{2}\sin\theta = \sin u$, so $\sqrt{2}\cos\theta\,\mathrm{d}\theta = \cos u\,\mathrm{d}u$, and u runs from 0 to $\frac{1}{2}\pi$ as θ runs from 0 to $\frac{1}{4}\pi$:

$$A = 2\int_0^{\pi/2} \cos u\cdot\frac{\cos u}{\sqrt{2}}\,\mathrm{d}u = \sqrt{2}\int_0^{\pi/2}\cos^2 u\,\mathrm{d}u = \sqrt{2}\cdot\frac{\pi}{4} = \frac{\pi}{2\sqrt{2}},$$

which is the required result. (Equivalently, the substitution $s = \sqrt{2}\sin\theta$ turns the integral into $\sqrt{2}\int_0^1 \sqrt{1-s^2}\,\mathrm{d}s$, which is $\sqrt{2}$ times a quarter of the unit disc.)

Checks

- At $\theta = 0$, $r_\pm = 1 \pm 1$, giving the points $(2,0)$ and O, consistent with the sketch.
- At $\theta = 0.01$, $r_- = \cos 0.01 - \sqrt{\cos 0.02} \approx 5.0002\times 10^{-5}$, against $\frac{1}{2}\theta^2 = 5\times 10^{-5}$.
- Numerically, $\pi/(2\sqrt{2}) \approx 1.11072$, and a direct numerical integration of $\frac{1}{2}(r_+^2 - r_-^2)$ over $[0, \frac{1}{4}\pi]$ gives the same value to six decimal places.
- A rough bound: the upper half of C_2 lies inside the rectangle $[0,2]\times[0,0.81]$, of area about 1.62, and it contains the triangle with vertices $(\frac{1}{2},0)$, $(2,0)$, $(\frac{1}{2},\frac{1}{2})$, of area 0.375. The value 1.11 sits sensibly between these.

Common slips

- Sketching C_1 with a corner, or with the curve crossing $y = x$, at $(\frac{1}{2}, \frac{1}{2})$: the infinite $\mathrm{d}r/\mathrm{d}\theta$ means the curve touches the radial line there.
- Drawing a smooth rounded end at O instead of a cusp: $y^2 \approx 2x^3$ is the classic cusp shape.
- Computing $\int \frac{1}{2} r_+^2 \, \mathrm{d}\theta$ alone, which is the area out to C_1 and includes the hollow region inside the inner branch.

Beyond STEP

The curve $y^2 = 2x^3$ near the origin is a semicubical parabola, the standard example of a cusp. Cusps show up whenever a smooth parametrisation has velocity zero at some instant: here $(x, y) \approx (\frac{1}{2}\theta^2, \frac{1}{2}\theta^3)$ has zero derivative at $\theta = 0$. Classifying such singular points of algebraic curves is the start of algebraic geometry. Polar curves and their areas also lead naturally to Green's theorem, which expresses an enclosed area as a line integral round its boundary. For a geometric view of curves like these, try Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP III 2020, Question 7

Pure • Second Order Differential Equations

STEP III 2020, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Part (i) shows that a second-order linear equation can sometimes be *factorised* into two first-order linear equations, each of which we can solve with an integrating factor. In part (ii) the work is to find the factorisation: matching coefficients gives a non-linear first-order equation for g, and a power-law guess solves it. After that we just solve two first-order equations in turn, using the initial conditions as we go.

Solution

Part (i)

The second equation gives u in terms of y:

$$u = \frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{g}(x)\, y.$$

Differentiating, using the product rule on $\mathrm{g}(x)y$,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \mathrm{g}'(x)\, y + \mathrm{g}(x)\frac{\mathrm{d}y}{\mathrm{d}x}.$$

Substituting both into $\dfrac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{f}(x)u = \mathrm{h}(x)$:

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + \mathrm{g}'(x)\, y + \mathrm{g}(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{f}(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{f}(x)\mathrm{g}(x)\, y = \mathrm{h}(x),$$

and collecting the $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ terms and the y terms gives exactly equation (1).

Part (ii)

An equation for g. Comparing (2) with (1), we need

$$\mathrm{g}(x) + \mathrm{f}(x) = 1 + \frac{4}{x}, \qquad \mathrm{g}'(x) + \mathrm{f}(x)\mathrm{g}(x) = \frac{2}{x} + \frac{2}{x^2}, \qquad \mathrm{h}(x) = 4x + 12.$$

The first gives $\mathrm{f} = 1 + \frac{4}{x} - \mathrm{g}$; putting this into the second eliminates f:

$$\mathrm{g}'(x) + \Big(1 + \frac{4}{x}\Big)\mathrm{g}(x) - \big(\mathrm{g}(x)\big)^2 = \frac{2}{x} + \frac{2}{x^2}.$$

This is the required first-order equation for g.

A power-law solution. With $\mathrm{g}(x) = kx^n$ the left-hand side is

$$knx^{n-1} + kx^n + 4kx^{n-1} - k^2x^{2n}.$$

? **Ask yourself** How do we choose n before we know k?

Look at the powers of x. The right-hand side contains only x^{-1} and x^{-2}, so the powers x^n, x^{n-1} and x^{2n} must only produce these. The choice $n = -1$ does it: $x^n = x^{-1}$, while x^{n-1} and x^{2n} are both x^{-2}. The left-hand side becomes

$$\frac{k}{x} + \frac{-k + 4k - k^2}{x^2} = \frac{k}{x} + \frac{3k - k^2}{x^2}.$$

Matching the x^{-1} terms gives $k = 2$, and then $3k - k^2 = 6 - 4 = 2$, which matches the x^{-2} term too. So

$$n = -1, \quad k = 2: \qquad \mathrm{g}(x) = \frac{2}{x}, \qquad \mathrm{f}(x) = 1 + \frac{4}{x} - \frac{2}{x} = 1 + \frac{2}{x}.$$

Solving (2). By part (i), (2) is now equivalent to the pair

$$\frac{\mathrm{d}u}{\mathrm{d}x} + \Big(1 + \frac{2}{x}\Big)u = 4x + 12, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2}{x}y = u.$$

Since the conditions are given at $x = 1$, we work on $x > 0$.

The equation for u. The integrating factor is $\exp\int\big(1 + \frac{2}{x}\big)\mathrm{d}x = \mathrm{e}^{x + 2\ln x} = x^2\mathrm{e}^x$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(x^2\mathrm{e}^x u\big) = \big(4x^3 + 12x^2\big)\,\mathrm{e}^x.$$

The right-hand side is exactly the derivative of $4x^3\mathrm{e}^x$ (product rule), so

$$x^2\mathrm{e}^x u = 4x^3\mathrm{e}^x + C, \qquad u = 4x + \frac{C\mathrm{e}^{-x}}{x^2}.$$

The value of u at $x = 1$ comes from the initial conditions: $u = y' + \frac{2}{x}y = -3 + 2 \times 5 = 7$. So $4 + C\mathrm{e}^{-1} = 7$, giving $C = 3\mathrm{e}$ and

$$u = 4x + \frac{3\mathrm{e}^{1-x}}{x^2}.$$

The equation for y. The integrating factor is $\mathrm{e}^{2\ln x} = x^2$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(x^2y\big) = x^2u = 4x^3 + 3\mathrm{e}^{1-x}, \qquad \text{hence} \qquad x^2y = x^4 - 3\mathrm{e}^{1-x} + D.$$

At $x = 1$: $5 = 1 - 3 + D$, so $D = 7$. Therefore

$$\boxed{y = x^2 + \frac{7 - 3\mathrm{e}^{1-x}}{x^2}}.$$

We used $y(1) = 5$ here and $y'(1) = -3$ in finding C, so both conditions hold.

Checks

- At $x = 1$: $y = 1 + 7 - 3 = 5$. Differentiating, $y' = 2x - 14x^{-3} + 3\mathrm{e}^{1-x}x^{-2} + 6\mathrm{e}^{1-x}x^{-3}$, which at $x = 1$ is $2 - 14 + 3 + 6 = -3$.
- The particular part $y = x^2$ should satisfy (2) on its own: $2 + (1 + \frac{4}{x})2x + (\frac{2}{x} + \frac{2}{x^2})x^2 = 2 + 2x + 8 + 2x + 2 = 4x + 12$. It does.
- Substituting the boxed y into (2) at many points between $x = 0.3$ and $x = 5$ by computer, the two sides agree to within the numerical differentiation error (about 10^{-5}).

Common slips

- Using $y'(1) = -3$ directly as the initial value for u. The initial value for u is $y' + \mathrm{g}y = 7$, not -3.
- Forgetting the $\mathrm{g}'(x)$ term in part (i): g depends on x, so $\mathrm{g}y$ must be differentiated with the product rule.
- Dropping a constant of integration in one of the two stages. There are two constants, which matches the two initial conditions.

Beyond STEP

This is the idea of factorising a differential operator: $(D + \mathrm{f})(D + \mathrm{g})y = \mathrm{h}$, where D means $\mathrm{d}/\mathrm{d}x$. Because g depends on x, the two factors don't commute, and that is why the cross term g' appears. The equation we found for g is a *Riccati equation*. Riccati equations are non-linear, but one known solution is enough to reduce the linear second-order problem to two first-order ones, as here. Boyce and DiPrima, *Elementary Differential Equations*, covers integrating factors, reduction of order and Riccati equations.

WORKED SOLUTION

STEP III 2020, Question 8

Pure • Sequences and Descent

STEP III 2020, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.8, *Counting the Rationals*: https://beyondstep.keithmcnulty.org/ch01/counting-rationals.html

The key idea

Every term with a large subscript is built from terms with about half that subscript. So any claim about the pair (u_n, u_{n+1}) can be pushed back to a pair with a smaller subscript, and an argument by *descent* (equivalently, choosing a smallest counterexample) does the work in every part. Writing out the first few terms, and how each consecutive pair produces two new ones, makes the pattern visible before any proof.

Solution

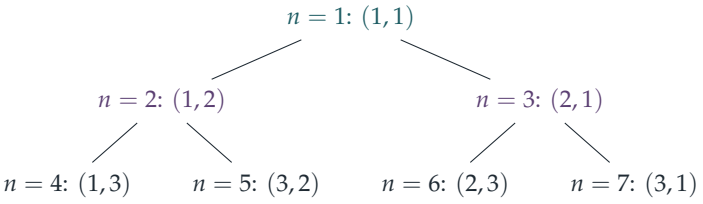
The first sixteen terms are

$$1,\ 1,\ 2,\ 1,\ 3,\ 2,\ 3,\ 1,\ 4,\ 3,\ 5,\ 2,\ 5,\ 3,\ 4,\ 1.$$

Every term is a positive integer: this holds for u_1, and each later term is either an earlier term or the sum of two earlier terms, so it follows by (strong) induction.

It is worth recording how a pair produces later pairs. If $(u_k, u_{k+1}) = (a, b)$, then directly from the definition

$$(u_{2k}, u_{2k+1}) = (a,\ a+b), \qquad (u_{2k+1}, u_{2k+2}) = (a+b,\ b). \tag{$*$}$$



The pair at position n is (u_n, u_{n+1}); its left child is at $2n$ and its right child at $2n+1$, by $(*)$. Every pair at position $n \geq 2$ arises in exactly this way from the pair at position $\lfloor n/2 \rfloor$.

Part (i)

Take a consecutive pair (u_n, u_{n+1}) with $n \geq 2$, and let $k = \lfloor n/2 \rfloor \geq 1$.

- If $n = 2k$, the pair is $(u_k,\ u_k + u_{k+1})$. The odd-subscript term $u_{2k+1} = u_k + u_{k+1}$ exceeds the even-subscript term $u_{2k} = u_k$ because $u_{k+1} \geq 1$.
- If $n = 2k+1$, the pair is $(u_k + u_{k+1},\ u_{k+1})$. Now $u_{2k+1} = u_k + u_{k+1}$ exceeds $u_{2k+2} = u_{k+1}$ because $u_k \geq 1$.

So in every pair except the first, $(u_1, u_2) = (1, 1)$, the term with odd subscript is strictly larger. In particular, *the two terms of a pair are equal only for the first pair.*

Part (ii)

Suppose $d > 1$ divides both u_n and u_{n+1}. This cannot happen for $n = 1$, since $u_1 = u_2 = 1$, so $n \geq 2$; let $k = \lfloor n/2 \rfloor$, so $1 \leq k < n$.

- If $n = 2k$: $d \mid u_{2k} = u_k$ and $d \mid u_{2k+1} = u_k + u_{k+1}$, so d divides the difference u_{k+1}.
- If $n = 2k+1$: $d \mid u_{2k+2} = u_{k+1}$ and $d \mid u_{2k+1} = u_k + u_{k+1}$, so $d \mid u_k$.

Either way d divides both u_k and u_{k+1}, an earlier consecutive pair, which is what we had to show.

Now suppose some consecutive pair has a common factor $d > 1$, and choose such a pair (u_n, u_{n+1}) with n as small as possible. As above $n \neq 1$, and the argument just given produces the pair at position $k < n$ with the same common factor, contradicting the minimality of n. Hence any two consecutive terms are co-prime.

Part (iii)

Suppose some ordered pair (a, b) occurs at two different positions, $(u_m, u_{m+1}) = (u_n, u_{n+1}) = (a, b)$ with $m < n$. Among all such repetitions, choose one with m as small as possible.

If $m = 1$ then $a = b = 1$; but $n \geq 2$, and by part (i) the pair at position n has unequal terms. So $m \geq 2$, and then also $a \neq b$ by part (i).

> **Ask yourself** How can we tell, just from a and b, whether the positions are odd or even?

Part (i) tells us that the larger term of the pair has the odd subscript, so the parity of the position is decided by the pair itself:

- If $a < b$, the second term has odd subscript, so m and n are both even: $m = 2j$,

$n = 2k$ with $j < k$. Then $u_j = u_{2j} = a$ and $u_{j+1} = u_{2j+1} - u_j = b - a$, and similarly $(u_k, u_{k+1}) = (a, b-a)$.

- If $a > b$, then m and n are both odd: $m = 2j+1$, $n = 2k+1$ with $1 \le j < k$. Then $u_{j+1} = u_{2j+2} = b$ and $u_j = u_{2j+1} - u_{j+1} = a - b$, and similarly $(u_k, u_{k+1}) = (a-b, b)$.

In both cases one ordered pair appears at the two different positions j and k, and $j \le m/2 < m$. This contradicts the choice of m. So no ordered pair of positive integers appears consecutively at two different places.

Part (iv)

Let $a > b$, with a, b co-prime, and suppose (a, b) is not a consecutive pair of the sequence. Then $a - b$ and b are positive integers, and they are co-prime: any common factor of $a - b$ and b also divides $(a - b) + b = a$, so it is a common factor of a and b and must be 1.

If $a - b$ were followed by b somewhere, say $(u_k, u_{k+1}) = (a - b, b)$, then by $(*)$

$$(u_{2k+1}, u_{2k+2}) = \big((a-b) + b,\ b\big) = (a, b),$$

contradicting the assumption on (a, b). So $(a - b, b)$ is also a co-prime pair that does not occur consecutively, and its sum is $a < a + b$.

The similar result for $a < b$: *a and $b - a$ are co-prime positive integers which do not occur consecutively with $b - a$ following a, and their sum b is smaller than $a + b$.* The proof is the same: a common factor of a and $b - a$ divides $a + (b - a) = b$, and if $(u_k, u_{k+1}) = (a, b - a)$ then $(u_{2k}, u_{2k+1}) = (a,\ a + (b - a)) = (a, b)$.

Part (v)

Since every term is a positive integer, $f(n) = u_n/u_{n+1}$ is a positive rational, so the range is contained in the positive rationals. We show that every positive rational is attained, and attained only once.

Every positive rational is in the range. Write a positive rational in lowest terms as p/q, with p, q co-prime positive integers. We claim every co-prime pair (p, q) occurs as (u_n, u_{n+1}) for some n. If not, among the co-prime pairs that do not occur choose one, (a, b), with $a + b$ as small as possible. We cannot have $a = b$: co-prime then forces $a = b = 1$, and $(1, 1)$ occurs at $n = 1$. So $a > b$ or $a < b$, and part (iv) supplies a co-prime pair that does not occur and has smaller sum. This contradicts minimality, so every co-prime pair (p, q) occurs, say at position n, and then $f(n) = p/q$.

f is one-to-one. Suppose $f(m) = f(n)$. By part (ii), u_m/u_{m+1} and u_n/u_{n+1} are both fractions in lowest terms with positive denominators. A positive rational has only one such representation, so $u_m = u_n$ and $u_{m+1} = u_{n+1}$. By part (iii) the same ordered pair cannot occur at two different places, so $m = n$.

Thus f is a bijection from the positive integers onto the positive rationals, and so it has an inverse f^{-1}: given p/q in lowest terms, $f^{-1}(p/q)$ is the unique position where p is immediately followed by q. For example, 3 is followed by 5 only at position 10, so

$f^{-1}(\frac{3}{5}) = 10$.

Checks

- The listed terms give pairs $(1,1), (1,2), (2,1), (1,3), (3,2), (2,3), (3,1), (1,4), \ldots$: odd-subscript terms are larger after the first pair, all pairs are co-prime, and no pair repeats.
- Running the descent of part (iv) on $3/5$: $(3,5) \to (3,2) \to (1,2) \to (1,1)$. Climbing back with $(*)$, $(1,1)$ at 1 gives $(1,2)$ at 2, $(3,2)$ at 5 and $(3,5)$ at 10; indeed $u_{10} = 3$, $u_{11} = 5$.
- A computer check over the first two million terms confirmed parts (i)–(iii), and found every co-prime pair (p,q) with $p+q \leq 21$ exactly once.

Common slips

- Forgetting to rule out the first pair $(1,1)$ separately in parts (ii) and (iii): it is the base case of each descent.
- In part (iii), assuming the two positions have the same parity without justifying it; part (i) is exactly what forces this.
- In part (v), proving only that the range is all positive rationals. An inverse also needs f to be one-to-one, which uses parts (ii) and (iii) together.

Beyond STEP

This is Stern's diatomic sequence, and listing u_n/u_{n+1} gives the Calkin–Wilf enumeration of the rationals: an explicit bijection between $\mathbb{N}$ and $\mathbb{Q}^+$, showing concretely that the rationals are countable. The descent in parts (iv) and (v) is the subtractive form of Euclid's algorithm, and the path from $(1,1)$ to (p,q) in the tree encodes the continued fraction of p/q. Davenport's *The Higher Arithmetic* covers Euclid's algorithm and continued fractions beautifully.

WORKED SOLUTION

STEP III 2020, Question 9

Mechanics • Rigid Body Equilibrium

STEP III 2020, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

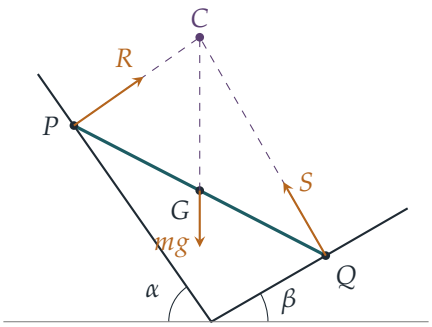
www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The rod is held by exactly three forces: its weight at the centre, and one contact force at each end. A body in equilibrium under three non-parallel forces has their lines of action meeting at a single point, so the whole question becomes geometry: find the height of that meeting point in two different ways and equate them. In part (ii) the normal reaction and the limiting friction at Q combine into one total reaction tilted at the angle of friction λ, where $\tan\lambda = \mu$, and the same geometry applies with β replaced by $\beta + \lambda$.

Solution

Model the rod as uniform and rigid, of length $2l$ with centre G, and the planes as fixed. Put Π_1 on the left and Π_2 on the right of the line where they meet, so P is on the left, higher, and Q is on the right, lower.



Part (i)

Both planes are smooth, so the reaction R at P is perpendicular to Π_1 and the reaction S at Q is perpendicular to Π_2. A line perpendicular to a plane inclined at α makes

angle α with the vertical; so the line of R rises from P leaning towards Q at angle α to the upward vertical, and the line of S rises from Q leaning towards P at angle β to the upward vertical.

The rod is in equilibrium under three forces, R, S and the weight mg acting vertically through G. Taking moments about the point C where the lines of R and S cross, R and S have no moment, so the weight must have none either: its line passes through C. So C lies vertically above G.

Ask yourself How do we turn "C is directly above G" into an equation?

Compute the height of C above Q along two routes. Since G is the midpoint and the rod makes angle θ with the horizontal, G is a horizontal distance $l\cos\theta$ from each end, and P is $2l\sin\theta$ higher than Q.

- Along the line of R: moving a horizontal distance $l\cos\theta$ along a line at angle α to the vertical raises us by $l\cos\theta\cot\alpha$. So C is $2l\sin\theta + l\cos\theta\cot\alpha$ above Q.
- Along the line of S: in the same way, C is $l\cos\theta\cot\beta$ above Q.

Equating, and dividing by $l\cos\theta \neq 0$,

$$2\tan\theta + \cot\alpha = \cot\beta, \qquad \text{that is} \qquad 2\tan\theta = \cot\beta - \cot\alpha,$$

as required. Since $\beta < \alpha$ (both acute), $\cot\beta > \cot\alpha$, so $\theta > 0$, consistent with P being higher than Q.

Part (ii)

Now Π_1 is still smooth, so the reaction at P is exactly as before. At Q there is a normal reaction N and a friction force F along Π_2.

Direction of friction. The rod is rigid, so the velocities of P and Q have equal components along the rod. If P starts to slide down Π_1 with speed u and Q moves up Π_2 with (signed) speed w, then resolving along the rod from P to Q, which points at angle ϕ below the horizontal,

$$u\cos(\alpha - \phi) = w\cos(\beta + \phi).$$

Here $\cos(\alpha - \phi) > 0$, and we shall see below that $\phi + \beta < \frac{1}{2}\pi$, so $w > 0$: Q is about to move *up* Π_2. Friction opposes this, so F acts down Π_2, and as equilibrium is limiting, $F = \mu N$.

Total reaction. Combine N and F into a single force S'. It makes angle λ with the normal, where $\tan\lambda = F/N = \mu$, and because F points down the plane (towards the line where the planes meet), it tilts the reaction further towards P. So the line of S' rises from Q at angle $\beta + \lambda$ to the upward vertical, leaning towards P. (If $\beta + \lambda > \frac{1}{2}\pi$ the line actually descends; the height formula below still holds, with a negative cotangent.)

The rod is again in equilibrium under three forces, so their lines meet at a point above G, and the argument of part (i) goes through word for word with θ replaced by ϕ and β

by $\beta+\lambda$:

$$2\tan\phi = \cot(\beta+\lambda) - \cot\alpha.$$

This shows $2\tan\phi < \cot\beta$, so $\tan\phi < \cot\beta = \tan(\frac{1}{2}\pi - \beta)$ and $\phi + \beta < \frac{1}{2}\pi$, which confirms the direction of friction we assumed.

Subtracting from $2\tan\theta = \cot\beta - \cot\alpha$,

$$2(\tan\theta - \tan\phi) = \cot\beta - \cot(\beta+\lambda) = \frac{\sin(\beta+\lambda)\cos\beta - \cos(\beta+\lambda)\sin\beta}{\sin\beta\,\sin(\beta+\lambda)}$$
$$= \frac{\sin\lambda}{\sin\beta\,(\sin\beta\cos\lambda + \cos\beta\sin\lambda)}.$$

Divide top and bottom by $\cos\lambda$ and use $\tan\lambda = \mu$, then take out $\cos\beta$:

$$2(\tan\theta - \tan\phi) = \frac{\mu}{\sin\beta\,(\sin\beta + \mu\cos\beta)} = \frac{\mu}{\sin\beta\cos\beta\,(\tan\beta+\mu)} = \frac{2\mu}{(\mu+\tan\beta)\sin 2\beta}.$$

Halving,

$$\tan\theta - \tan\phi = \frac{\mu}{(\mu+\tan\beta)\sin 2\beta},$$

which is the required result.

Checks

- With $\mu = 0$ the right-hand side is 0, so $\phi = \theta$: part (ii) reduces to part (i), as it should.
- The right-hand side is positive, so $\phi < \theta$: friction lets the rod rest flatter than it could on smooth planes, which makes sense as P is the end that is about to slide down.
- Take $\alpha = 60°$, $\beta = 30°$. Then $2\tan\theta = \sqrt{3} - 1/\sqrt{3} = 2/\sqrt{3}$, so $\theta = 30°$. With $\mu = 0.5$ the formula gives $\tan\phi = 0.5774 - 0.5/(1.0774 \times 0.8660) \approx 0.0415$, so $\phi \approx 2.4°$. Solving the equations of horizontal and vertical resolution and moments about G directly for these values (a computer check, repeated for thousands of random angles and coefficients) gives the same ϕ.
- Any μ can be tested quickly via $2\tan\phi = \cot(\beta+\lambda) - \cot\alpha$: here $\lambda \approx 26.57°$, $\cot 56.57° \approx 0.6606$, and $(0.6606 - 0.5774)/2 \approx 0.0416$, agreeing to rounding.

Common slips

- Putting friction up Π_2. When P slides down, the rod's length forces Q *up* its plane, so friction at Q acts down it.
- Measuring the reactions' angles from the horizontal instead of the vertical: the normal to a plane at angle α is at angle α to the *vertical*.

Beyond STEP

The three-force concurrency used here is a special case of the general conditions for rigid-body equilibrium: zero resultant force and zero resultant moment about any point. Combining friction and normal reaction into a "cone of friction" of half-angle λ is how engineers think about stability of ladders, wedges and screws, and the velocity argument for the direction of friction is the idea of an instantaneous centre of rotation. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction in this spirit.

WORKED SOLUTION

STEP III 2020, Question 10

Mechanics • Simple Harmonic Motion

STEP III 2020, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Hanging from a spring, the particle obeys simple harmonic motion about its equilibrium position, and that fixes Ω in terms of k, g and a. On the platform, the particle's acceleration is the platform's, which is known in terms of x, so Newton's second law gives the reaction directly. Contact needs the reaction to be non-negative all the time; since it turns out to be *linear* in x, we only have to check it at the two ends of the platform's travel, and which end matters depends on whether ω is smaller or larger than Ω.

Solution

Model the particle as a point mass and the spring as light, obeying Hooke's law in both extension and compression. Throughout, take upwards as positive for forces on the particle.

The hanging particle. Let y be the distance of the particle below A, so the extension is $y-a$ and the tension is $kmg(y-a)/a$, acting upwards on the particle. Newton's second law, measured downwards, gives

$$m\ddot{y} = mg - \frac{kmg}{a}(y-a) \qquad \Longrightarrow \qquad \ddot{y} = -\frac{kg}{a}\left(y - a - \frac{a}{k}\right).$$

With $z = y - a(1+\frac{1}{k})$ this is $\ddot{z} = -\frac{kg}{a}z$, which is simple harmonic motion about the equilibrium point $y = a(1+\frac{1}{k})$. The particle is released from rest away from this point (any extension other than a/k gives $z \neq 0$), so it oscillates vertically with angular frequency $\sqrt{kg/a}$ and period $2\pi\sqrt{a/(kg)}$. Setting this equal to $2\pi/\Omega$ gives $\Omega^2 = kg/a$, that is,

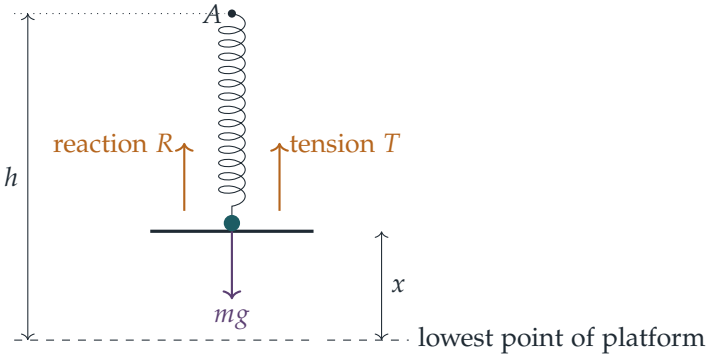
$$kg = a\Omega^2.$$

We use this in the form $g/\Omega^2 = a/k$ below.

The platform. The platform performs simple harmonic motion of amplitude b and angular frequency ω, so its height x above its lowest point ranges over $0 \le x \le 2b$, with centre $x = b$. Hence its acceleration is

$$\ddot{x} = -\omega^2(x - b) = \omega^2(b - x).$$

(For instance, $x = b(1 - \cos\omega t)$ gives $\ddot{x} = \omega^2 b\cos\omega t = \omega^2(b - x)$.)



The reaction. While the particle is in contact, it moves with the platform, so its height above the platform's lowest point is x and its acceleration is $\ddot{x}$. Its distance below A is $h - x$, so the extension of the spring is $h - x - a$ and the tension is

$$T = \frac{kmg}{a}(h - x - a) = m\Omega^2(h - x - a),$$

using $kg/a = \Omega^2$. (If $h - x < a$ the spring is compressed and T is negative, a thrust, and the same expression still holds.) The forces on the particle are R and T upwards and mg downwards, so

$$R + T - mg = m\ddot{x} = m\omega^2(b - x),$$

and therefore

$$R = mg - m\Omega^2(h - x - a) + m\omega^2(b - x) = mg + m\Omega^2(a + x - h) + m\omega^2(b - x),$$

as required.

Staying in contact. The particle stays on the platform throughout exactly when $R \ge 0$ for every x in $[0, 2b]$. Collecting the terms in x,

$$\frac{R}{m} = g + \Omega^2(a - h) + \omega^2 b + (\Omega^2 - \omega^2)\,x.$$

? **Ask yourself** Why is it enough to look at only one value of x?

R is a linear function of x, so over the interval $0 \le x \le 2b$ its least value is at one end: at $x = 0$ if the slope $\Omega^2 - \omega^2$ is positive, and at $x = 2b$ if it is negative. The platform reaches both ends in every cycle, so $R \ge 0$ throughout is equivalent to $R \ge 0$ at the appropriate end.

Case $\omega < \Omega$. The least value is at $x = 0$ (the bottom of the motion), so we need

$$g + \Omega^2(a - h) + \omega^2 b \ge 0 \iff h \le a + \frac{g}{\Omega^2} + \frac{\omega^2 b}{\Omega^2} = a\left(1 + \frac{1}{k}\right) + \frac{\omega^2 b}{\Omega^2},$$

dividing by $\Omega^2 > 0$ and using $g/\Omega^2 = a/k$. This is the required inequality.

Case $\omega > \Omega$. Now the least value is at $x = 2b$ (the top of the motion), so we need

$$g + \Omega^2(a - h) + \omega^2 b + 2b(\Omega^2 - \omega^2) \ge 0 \iff h \le a\left(1 + \frac{1}{k}\right) + b\left(2 - \frac{\omega^2}{\Omega^2}\right).$$

A bound for every ω. Suppose the particle stays in contact.

- If $\omega < \Omega$, then $\omega^2 b/\Omega^2 < b$ (as $b > 0$), so $h \le a(1 + \frac{1}{k}) + \omega^2 b/\Omega^2 < a(1 + \frac{1}{k}) + b$.
- If $\omega > \Omega$, then $2 - \omega^2/\Omega^2 < 1$, so $h \le a(1 + \frac{1}{k}) + b(2 - \omega^2/\Omega^2) < a(1 + \frac{1}{k}) + b$.
- If $\omega = \Omega$, the x terms in R cancel and $R/m = g + \Omega^2(a - h) + \Omega^2 b$ is constant; $R \ge 0$ gives exactly $h \le a(1 + \frac{1}{k}) + b$.

So in every case

$$h \le a\left(1 + \frac{1}{k}\right) + b$$

is necessary, whatever the value of ω. Since $h - b$ is the depth of the centre of the platform's motion below A, and $a(1 + \frac{1}{k})$ is the depth of the hanging equilibrium, this says: the platform must oscillate about a point no lower than the particle's natural equilibrium position.

Checks

- Stepping the free motion numerically with $a = 1$, $k = 2$, $g = 9.8$ gives a period with $a\Omega^2 \approx 19.600$, matching $kg = 19.6$.
- If the platform is at rest ($b = 0$), both bounds become $h \le a(1 + \frac{1}{k})$: the platform must not be below the equilibrium point, or the spring would lift the particle off. That is exactly right.
- At $\omega = \Omega$ the two case bounds agree, both giving $a(1 + \frac{1}{k}) + b$, as they should by continuity.

Common slips

- Checking only the top of the motion. For slow platforms ($\omega < \Omega$) the particle is most at risk at the *bottom*, where the spring pulls up hardest.

- Forgetting that the spring can be compressed: the tension formula $kmg(h-x-a)/a$ is valid with either sign, so no separate case is needed.

Beyond STEP

This is a forced oscillator in disguise: the spring tries to make the particle oscillate at its natural frequency Ω, while the platform imposes frequency ω, and the reaction is the force needed to reconcile the two. Undergraduate courses on oscillations study driven systems, resonance and the way the response depends on ω/Ω. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats these ideas carefully.

WORKED SOLUTION

STEP III 2020, Question 11

Probability and Statistics • Functions of Random Variables

STEP III 2020, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

A strictly decreasing function reverses inequalities, and because f is its own inverse we can apply it to both sides of $\mathrm{f}(X) \leq y$ to get an event about X alone. The expectation then comes from integrating by parts, which swaps f' for f. In part (ii) the relation between Z and X is symmetric in the two variables, so $Z = \mathrm{f}(X)$ for a self-inverse f and part (i) applies. The inequality is then just the statement that a variance is positive.

Solution

Part (i)

Since f^{-1} is defined on all of $[a, b]$, f is a bijection of $[a, b]$ onto itself, so Y takes values in $[a, b]$. Fix $y \in [a, b]$. Because f is strictly decreasing, applying it to both sides of an inequality reverses the inequality, and $\mathrm{f}(\mathrm{f}(x)) = x$. Hence, for $x \in [a, b]$,

$$\mathrm{f}(x) \leq y \iff \mathrm{f}(\mathrm{f}(x)) \geq \mathrm{f}(y) \iff x \geq \mathrm{f}(y).$$

(The reverse implication holds in the same way, by applying f to $x \geq \mathrm{f}(y)$.) Since X is uniform on $[a, b]$ and $\mathrm{f}(y) \in [a, b]$,

$$\mathrm{P}(Y \leq y) = \mathrm{P}\big(X \geq \mathrm{f}(y)\big) = \frac{b - \mathrm{f}(y)}{b - a},$$

as required.

Differentiating, the probability density function of Y is

$$g(y) = -\frac{\mathrm{f}'(y)}{b - a} \quad \text{for } a \leq y \leq b,$$

and zero otherwise. This is non-negative, since f is decreasing. Then, integrating by

parts with $u = y^2$ and $\mathrm{d}v = -\mathrm{f}'(y)\,\mathrm{d}y$,

$$\begin{aligned}\mathrm{E}(Y^2) &= \frac{1}{b-a}\int_a^b y^2\big(-\mathrm{f}'(y)\big)\,\mathrm{d}y \\ &= \frac{1}{b-a}\Big[-y^2\mathrm{f}(y)\Big]_a^b + \int_a^b \frac{2y\,\mathrm{f}(y)}{b-a}\,\mathrm{d}y \\ &= \frac{-b^2a + a^2b}{b-a} + \int_a^b \frac{2y\,\mathrm{f}(y)}{b-a}\,\mathrm{d}y,\end{aligned}$$

using $\mathrm{f}(b) = a$ and $\mathrm{f}(a) = b$. The first term is $-ab(b-a)/(b-a) = -ab$. Renaming the dummy variable y as x gives

$$\mathrm{E}(Y^2) = -ab + \int_a^b \frac{2x\,\mathrm{f}(x)}{b-a}\,\mathrm{d}x.$$

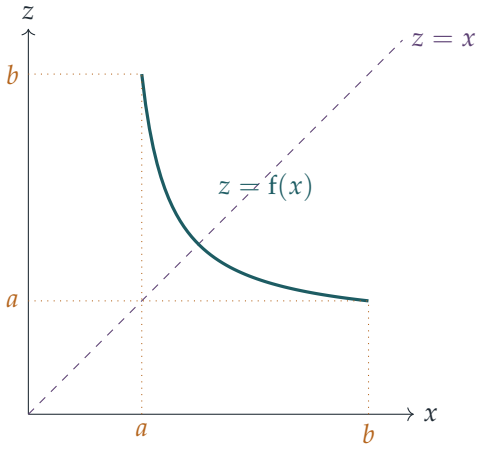
Part (ii)

Note that $\frac{1}{c} = \frac{1}{a} + \frac{1}{b}$ gives $c = \dfrac{ab}{a+b}$, and $c < a$ because $\frac{1}{c} > \frac{1}{a}$. Solving for Z,

$$\frac{1}{Z} = \frac{1}{c} - \frac{1}{X} = \frac{X-c}{cX}, \qquad \text{so} \qquad Z = \mathrm{f}(X) \ \text{ where } \ \mathrm{f}(x) = \frac{cx}{x-c}.$$

Since $x \geq a > c$, the denominator is positive on $[a,b]$. We check that f satisfies the three conditions of part (i).

- $\frac{1}{\mathrm{f}(a)} = \frac{1}{c} - \frac{1}{a} = \frac{1}{b}$, so $\mathrm{f}(a) = b$. Similarly $\frac{1}{\mathrm{f}(b)} = \frac{1}{c} - \frac{1}{b} = \frac{1}{a}$, so $\mathrm{f}(b) = a$.
- $\frac{1}{\mathrm{f}(x)} = \frac{1}{c} - \frac{1}{x}$ is positive and strictly increasing on $[a,b]$, so $\mathrm{f}(x)$ is strictly decreasing there.
- The relation $\frac{1}{z} + \frac{1}{x} = \frac{1}{c}$ is symmetric in x and z. So $z = \mathrm{f}(x)$ exactly when $x = \mathrm{f}(z)$, which means $\mathrm{f} = \mathrm{f}^{-1}$.



The graph of f runs from (a,b) to (b,a) and is symmetric in the line $z = x$, as a self-inverse function must be.

? Ask yourself Why bother finding $\mathrm{E}(Z)$ as well as $\mathrm{E}(Z^2)$?

The inequality has to come from $\mathrm{Var}(Z) > 0$, and the variance needs both moments. The logarithm will appear in each moment, and its first power will cancel, leaving its square.

For both integrals, divide out:

$$\mathrm{f}(x) = c + \frac{c^2}{x-c}, \qquad x\,\mathrm{f}(x) = \frac{cx^2}{x-c} = c\Big(x + c + \frac{c^2}{x-c}\Big).$$

Write $L = \ln\Big(\dfrac{b-c}{a-c}\Big)$, which is positive because $b-c > a-c > 0$. Then

$$\mathrm{E}(Z) = \frac{1}{b-a}\int_a^b \Big(c + \frac{c^2}{x-c}\Big)\,\mathrm{d}x = c + \frac{c^2 L}{b-a}.$$

By part (i),

$$\begin{aligned}\mathrm{E}(Z^2) &= -ab + \frac{2c}{b-a}\int_a^b \Big(x + c + \frac{c^2}{x-c}\Big)\,\mathrm{d}x \\ &= -ab + \frac{2c}{b-a}\Big(\frac{b^2-a^2}{2} + c(b-a) + c^2 L\Big) \\ &= -ab + c(a+b) + 2c^2 + \frac{2c^3 L}{b-a}.\end{aligned}$$

Now $c(a+b) = ab$, so the first two terms cancel. Subtracting $\mathrm{E}(Z)^2 = c^2 + \dfrac{2c^3 L}{b-a} + \dfrac{c^4 L^2}{(b-a)^2}$, the terms in L cancel and we get

$$\mathrm{Var}(Z) = c^2 - \frac{c^4 L^2}{(b-a)^2}.$$

Z is not constant: by part (i) its distribution function $\big(b - \mathrm{f}(y)\big)/(b-a)$ increases strictly from 0 to 1 across $[a,b]$. So $\mathrm{Var}(Z) > 0$, which gives

$$L^2 < \frac{(b-a)^2}{c^2}.$$

Both L and $(b-a)/c$ are positive, so we may take square roots:

$$\ln\Big(\frac{b-c}{a-c}\Big) < \frac{b-a}{c}.$$

Checks

- In part (i), try the simplest self-inverse example $\mathrm{f}(x) = a + b - x$. Then Y is uniform on $[a,b]$, and the formula gives $\mathrm{E}(Y^2) = -ab + \frac{2}{b-a}\int_a^b x(a+b-x)\,\mathrm{d}x = \frac{1}{3}(a^2 + ab + b^2)$, which is the correct second moment of a uniform variable.

- With $a = 1$, $b = 3$ we have $c = \frac{3}{4}$ and $L = \ln 9 \approx 2.197$, while $(b-a)/c = \frac{8}{3} \approx 2.667$. The inequality holds, and $\mathrm{Var}(Z) = \frac{9}{16} - \frac{81}{256} \cdot \frac{(\ln 9)^2}{4} \approx 0.180$. Integrating f and f^2 numerically by computer gives the same variance.
- As $b \to a$, both sides of the inequality tend to 0, and $\mathrm{Var}(Z) \to 0$, as it should: Z is squeezed into $[a, b]$.

Common slips

- Forgetting that a decreasing function reverses the inequality, and so writing $\mathrm{P}(Y \leq y) = \mathrm{P}(X \leq \mathrm{f}(y))$.
- Leaving out the minus sign in the density $-\mathrm{f}'(y)/(b-a)$. A density cannot be negative.
- Not saying why $\mathrm{Var}(Z)$ is *strictly* positive, or taking square roots without noting that both sides are positive.

Beyond STEP

Part (i) is a special case of the transformation rule for densities: if $Y = \mathrm{f}(X)$ with f monotonic, then $g_Y(y) = g_X\big(\mathrm{f}^{-1}(y)\big)\,\big|(\mathrm{f}^{-1})'(y)\big|$. Getting a strict inequality from $\mathrm{Var}(Z) > 0$ is an example of a powerful trick: $\mathrm{E}(Z)^2 \leq \mathrm{E}(Z^2)$ is Jensen's inequality (or Cauchy–Schwarz) in disguise, and choosing a clever random variable proves many classical inequalities. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

WORKED SOLUTION

STEP III 2020, Question 12

Probability and Statistics • Geometric Distribution Independence

STEP III 2020, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Everything is a function of the pair (X, Y) of independent geometric variables, with $\mathrm{P}(X = x,\, Y = y) = p^2 q^{x+y-2}$. Sums and differences are best handled by adding up this joint probability along a line in the (x, y) grid, while maxima and minima are easiest through cumulative probabilities such as $\mathrm{P}(T \le t)$. For independence, look for a value of one variable that pins down the other completely.

Solution

Since A tosses until the first head, X takes the value $k \ge 1$ when the first $k-1$ tosses are tails and the kth is a head:

$$\mathrm{P}(X = k) = pq^{k-1}, \qquad k = 1, 2, 3, \ldots,$$

and the same holds for Y. The two players toss separately, so X and Y are independent, and

$$\mathrm{P}(X = x,\, Y = y) = p^2 q^{x+y-2} \qquad (x, y \ge 1).$$

Two facts we use repeatedly: $\mathrm{P}(X > k) = q^k$ (the first k tosses are all tails), so $\mathrm{P}(X \le k) = 1 - q^k$; and $1 - q^2 = (1-q)(1+q) = p(1+q)$.

Part (i)

The sum. S takes values $s \ge 2$. For $S = s$ we need $X = x$ and $Y = s - x$ with $1 \le x \le s-1$. Each of these $s-1$ outcomes has probability $p^2 q^{s-2}$, so

$$\mathrm{P}(S = s) = (s-1)p^2 q^{s-2}, \qquad s = 2, 3, 4, \ldots$$

The maximum. $T \le t$ exactly when both $X \le t$ and $Y \le t$, so by independence

$\mathrm{P}(T \le t) = (1-q^t)^2$. Hence, for $t \ge 1$,

$$\begin{aligned}\mathrm{P}(T = t) &= \mathrm{P}(T \le t) - \mathrm{P}(T \le t-1) = (1-q^t)^2 - (1-q^{t-1})^2 \\ &= \left[(1-q^t) - (1-q^{t-1})\right]\left[(1-q^t) + (1-q^{t-1})\right] \\ &= (q^{t-1} - q^t)(2 - q^{t-1} - q^t) \\ &= pq^{t-1}(2 - q^{t-1} - q^t),\end{aligned}$$

using $q^{t-1} - q^t = q^{t-1}(1-q) = pq^{t-1}$. This is the required result. (For $t = 1$ the formula gives $p(1-q) = p^2$, correctly.)

Part (ii)

The difference. For $U = 0$ we need $X = Y = k$ for some $k \ge 1$:

$$\mathrm{P}(U = 0) = \sum_{k=1}^{\infty} p^2 q^{2k-2} = \frac{p^2}{1-q^2} = \frac{p^2}{p(1+q)} = \frac{p}{1+q}.$$

For $u \ge 1$, either $Y = X + u$ or $X = Y + u$; these are disjoint and by symmetry equally likely, so

$$\mathrm{P}(U = u) = 2\sum_{k=1}^{\infty} \mathrm{P}(X = k,\, Y = k+u) = 2\sum_{k=1}^{\infty} p^2 q^{2k+u-2} = \frac{2p^2 q^u}{1-q^2} = \frac{2pq^u}{1+q}.$$

The minimum. $W > w$ exactly when both $X > w$ and $Y > w$, so $\mathrm{P}(W > w) = q^{2w}$ and, for $w \ge 1$,

$$\mathrm{P}(W = w) = \mathrm{P}(W > w-1) - \mathrm{P}(W > w) = q^{2w-2} - q^{2w} = (1-q^2)\, q^{2(w-1)}.$$

So W is itself geometric, with "success" probability $1 - q^2$: each round of one toss by A and one by B produces at least one head with probability $1 - q^2$.

Part (iii)

If $S = 2$ then, since $X, Y \ge 1$, we must have $X = Y = 1$, and then $T = 1$. So $S = 2$ and $T = 3$ cannot happen together:

$$\mathrm{P}(S = 2 \text{ and } T = 3) = 0.$$

On the other hand $\mathrm{P}(S = 2) = p^2 > 0$ and $\mathrm{P}(T = 3) = pq^2(2 - q^2 - q^3) > 0$, because $0 < q < 1$ gives $q^2 + q^3 < 2$. Hence $\mathrm{P}(S = 2) \times \mathrm{P}(T = 3) > 0$, and the two sides differ.

Part (iv)

U and W are independent. We must show $\mathrm{P}(U = u,\, W = w) = \mathrm{P}(U = u)\,\mathrm{P}(W = w)$ for every $u \ge 0$ and $w \ge 1$. Knowing both U and W tells us the smaller value is w and the larger is $w + u$.

Ask yourself Which pairs (X, Y) give $U = u$ and $W = w$?

If $u = 0$ the only pair is (w, w). If $u \geq 1$ there are exactly two: $(w,\ w+u)$ and $(w+u,\ w)$, each with probability p^2q^{2w+u-2}.

For $u = 0$, $\mathrm{P}(U = 0,\ W = w) = p^2q^{2w-2}$, while

$$\mathrm{P}(U = 0)\,\mathrm{P}(W = w) = \frac{p}{1+q} \cdot p(1+q)\,q^{2w-2} = p^2q^{2w-2}.$$

For $u \geq 1$, $\mathrm{P}(U = u,\ W = w) = 2p^2q^{2w+u-2}$, while

$$\mathrm{P}(U = u)\,\mathrm{P}(W = w) = \frac{2pq^u}{1+q} \cdot p(1+q)\,q^{2w-2} = 2p^2q^{2w+u-2}.$$

These agree in every case, so U and W are independent.

No other pair is independent. The remaining pairs are (S, T), (S, U), (S, W), (T, U) and (T, W). To show two variables are *not* independent, one pair of values with $\mathrm{P}(A = a,\ B = b) \neq \mathrm{P}(A = a)\,\mathrm{P}(B = b)$ is enough.

Ask yourself Which single value of S or T forces everything else?

The event $S = 2$ and the event $T = 1$ are the same event, $\{X = Y = 1\}$, of probability $p^2 > 0$; on it $T = 1$, $U = 0$ and $W = 1$. Every value of T, U and W in the list below has positive probability by the formulas in parts (i) and (ii), since $0 < p, q < 1$. So:

- (S, T): $\mathrm{P}(S = 2,\ T = 3) = 0 \neq \mathrm{P}(S = 2)\,\mathrm{P}(T = 3)$, by part (iii);
- (S, U): $\mathrm{P}(S = 2,\ U = 1) = 0$, but $\mathrm{P}(S = 2)\,\mathrm{P}(U = 1) > 0$;
- (S, W): $\mathrm{P}(S = 2,\ W = 2) = 0$, but $\mathrm{P}(S = 2)\,\mathrm{P}(W = 2) > 0$;
- (T, U): $\mathrm{P}(T = 1,\ U = 1) = 0$, but $\mathrm{P}(T = 1)\,\mathrm{P}(U = 1) > 0$;
- (T, W): $\mathrm{P}(T = 1,\ W = 2) = 0$, but $\mathrm{P}(T = 1)\,\mathrm{P}(W = 2) > 0$.

So U and W form the only independent pair among S, T, U, W.

Checks

- Each distribution sums to 1: for instance $\sum_{t\geq 1} \mathrm{P}(T = t)$ telescopes to $\lim(1 - q^t)^2 = 1$, and $\frac{p}{1+q} + \sum_{u\geq 1} \frac{2pq^u}{1+q} = \frac{p}{1+q}\left(1 + \frac{2q}{p}\right) = \frac{p+2q}{1+q} = 1$.
- With $p = q = \frac{1}{2}$: $\mathrm{P}(T = 2) = \frac{1}{4}\left(2 - \frac{1}{2} - \frac{1}{4}\right) = \frac{5}{16}$, which matches listing the three outcomes $(1,2), (2,1), (2,2)$, with probabilities $\frac{1}{8} + \frac{1}{8} + \frac{1}{16}$.
- With $p = q = \frac{1}{2}$, $\mathrm{P}(U = 1,\ W = 1) = \mathrm{P}\{(1,2), (2,1)\} = \frac{1}{4}$, and $\mathrm{P}(U = 1)\,\mathrm{P}(W = 1) = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$, as independence requires.
- All the formulas, and the independence of U and W, were confirmed by computer by summing the joint distribution for $p = 0.3$, 0.55 and 0.8.

Common slips

- Forgetting the factor 2 in $\mathrm{P}(U = u)$ for $u \geq 1$, or wrongly including it for $u = 0$, where $X = Y$ counts once.
- Trying to find $\mathrm{P}(T = t)$ by adding cases directly and double-counting $X = Y = t$; the cumulative approach avoids this.
- Thinking that checking a single pair of values proves independence: that needs *all* u and w, whereas one failure is enough for dependence.

Beyond STEP

The independence of U and W is the discrete form of the memoryless property: once the first head appears at time W, the other player's remaining wait is a fresh geometric variable, regardless of how long it took. The continuous analogue, for independent exponential variables, says the minimum and the gap to the maximum are independent, which underlies Poisson processes and queueing theory. Grimmett and Welsh, *Probability: An Introduction*, develops these ideas.

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